

Prefoldin 5 is a microtubule-associated protein that suppresses Tau-aggregation and neurotoxicity

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This work details the finding that in at least one of the subunits of the heterohexameric chaperone complex Pfdn5 has additional functions beyond its contribution to cytoskeletal protein folding in *Drosophila*. The authors provide **convincing** evidence that it is a hitherto unknown microtubule associated protein in addition to regulating microtubule organization and levels of tubulin monomers. The **important** findings show that Pfdn5 loss exaggerates pathological manifestations of mutant human Tau bearing FTDP-17 linked mutations in *Drosophila*, while its overexpression suppresses them, suggesting that the latter may constitute a future therapeutic approach.

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Abstract

Tauopathies represent a major class of neurodegenerative disorders associated with intracellular aggregates of the microtubule-associated protein Tau. To identify molecular modulators of Tau toxicity, we used a genetic screen to identify protein chaperones whose RNAi-mediated knockdown could modulate hTau^{V337M}-induced eye-ommatidial degeneration in *Drosophila*. This screen identified the Prefoldins Pfdn5 and Pfdn6 as strong modifiers of hTau^{V337M} cytotoxicity. Consistent with the known function of Pfdn as a cotranslational chaperone for tubulin, *Pfdn5* mutants showed substantially reduced levels of tubulin monomer. However, additional microtubule-related functions were indicated by the robust unexpected association of Pfdn5 with axonal microtubules *in vivo*, as well as binding with stabilized microtubules in biochemical assays. Loss of Pfdn5 resulted in neuromuscular junctions (NMJ) defects similar to those previously described in hTau-expressing flies: namely, increased supernumerary boutons and fewer microtubule loops within mature presynaptic boutons. Significantly, synaptic phenotypes caused by hTau^{V337M} overexpression were also strongly enhanced in a *Pfdn5* mutant background. Consistent with a role in modulating Tau toxicity, not only did loss of *Pfdn5* result in increased accumulations of Tau-aggregates in hTau^{V337M} expressing neurons, but also neuronal overexpression of Prefoldin

strikingly ameliorated age-dependent neurodegeneration and memory deficits induced by pathological hTau. Together, these and other observations described herein: (a) provide new insight into Prefoldin-microtubule interactions; (b) point to essential posttranslational roles for Pfdn5 in controlling Tau-toxicity *in vivo*; and (c) demonstrate that Pfdn5 overexpression is sufficient to restrict Tau-induced neurodegeneration.

Introduction

Aberrant accumulation of misfolded protein aggregates is associated with neuroinflammation, neuronal death, and progressive cognitive decline in diverse groups of neurodegenerative diseases (Ross and Poirier 2004 [↗](#); Sweeney et al. 2017 [↗](#); Soto and Pritzkow 2018 [↗](#); Lashuel 2021 [↗](#)). Alzheimer's disease, as well as a subset of other neurodegenerative disorders, together referred to as Tauopathies, are defined by accumulated intracellular aggregates of the Tubulin-associated unit (Tau) protein (Iqbal et al. 2010 [↗](#); Jouanne et al. 2017 [↗](#); Pinzi et al. 2023 [↗](#)). Despite substantial clinical interest and decades of research, effective therapeutic interventions for treating Tauopathies are still unavailable (Khanna et al. 2016 [↗](#); Lashuel 2021 [↗](#)).

Tau is predominantly expressed in neurons, where it stabilizes microtubules, thus facilitating intra-axonal transport (Avila et al. 2004 [↗](#); Kent et al. 2020 [↗](#); Robbins et al. 2021 [↗](#)). Several mutations in the Tau protein have been identified that contribute to a wide spectrum of Tauopathies, including Alzheimer's disease, Pick's disease, progressive supranuclear palsy (PSP), corticobasal degeneration (CBD), and frontotemporal dementia with Parkinsonism linked to chromosome 17 (FTDP-17) (Goedert and Jakes 2005 [↗](#); Wolfe 2009 [↗](#)). These pathogenic mutations enhance the propensity for Tau protein hyperphosphorylation at Ser/Thr residues, leading to the formation of neurofibrillary tangles via self-aggregation (Tzioras et al. 2023 [↗](#)). The phosphorylated Tau disengages from the microtubule, potentially altering axonal transport and contributing to synapse loss and/or axon retraction (Guo et al. 2020 [↗](#); Tzioras et al. 2023 [↗](#)). Thus, the self-aggregation of Tau and destabilization of microtubules may contribute to the progression of Tau pathogenesis. Such a model is supported by studies in *Drosophila* and rodent models of Tauopathies. Several of these models of Tauopathy show disrupted microtubules, synaptic abnormalities, and abnormal motor behavior (Stubbs et al. 2023 [↗](#)). Significantly, pharmacological stabilization of microtubules or reducing Tau levels can revert at least some of the defects observed in these Tauopathy models (Zhang et al. 2005 [↗](#); Brunden et al. 2010 [↗](#)). However, alternative approaches to mitigate Tau-induced neurodegeneration are required because the currently available microtubule-targeting drugs are toxic at concentrations required to have an effect in the brain (Yu et al. 2021 [↗](#)). One approach, suggested by several studies demonstrating a role for chaperone systems in Tauopathies (Perez et al. 1991 [↗](#); Renkawek et al. 1994 [↗](#); Ostapchenko et al. 2013 [↗](#)), is to identify and manipulate specific molecular chaperones that directly or indirectly control Tau aggregation and Tau-induced neurotoxicity *in vivo* (Blard et al. 2007 [↗](#); Darling et al. 2021 [↗](#)).

Molecular chaperones facilitate proper protein folding, prevent protein aggregation, and solubilize or facilitate autophagic or proteasomal elimination of protein aggregates (Warrick et al. 1999 [↗](#); Dou et al. 2003 [↗](#); Buchner 2019 [↗](#)). Consistent with this, enhanced expression of Hsp70 or HSP90 chaperones in mouse neuroblastoma N2A cells reduces pathological Tau levels by promoting the partitioning of Tau onto microtubules (Dou et al. 2003 [↗](#)). On the other hand, as chaperones stabilize misfolded protein states, the expression of certain chaperones or cochaperones can sometimes also promote and facilitate the aggregation of Tau (Bhattacharya et al. 2020 [↗](#); Criado-Marrero et al. 2021 [↗](#)). For instance, expression of HSP90 cochaperones, FKBP52, or Aha1 in the mouse brain enhances Tau aggregation, neuroinflammation, and cognitive decline in the Tau transgenic mouse model (Shelton et al. 2017 [↗](#); Criado-Marrero et al. 2021 [↗](#)). These and other data indicate that: (a) chaperones not only alleviate but also aggravate Tau aggregation and hence identification and analysis of chaperones that modulate Tau-aggregation

and toxicity are required to understand biological and pathogenic mechanisms involved in Tauopathy, and (b) genetic or pharmacological manipulation of specific chaperone activities could be of possible therapeutic value for treating Tau-induced neurodegeneration.

Here, we report that Pfdn5 colocalizes with axonal microtubules and physically associates with stable microtubules. Loss of Pfdn5 resulted in a remarkable reduction in tubulin levels, disrupting microtubules in otherwise wild-type *Drosophila*, as well as the aggregation of Tau in axons and larval brain of the *Drosophila* hTau^{V337M} disease model. Moreover, Pfdn5 deletion exacerbates Tau-induced neurotoxicity, overexpression of Pfdn5 mitigates the age-dependent progression of neurodegeneration and suppresses the learning and long-term memory deficits associated with Tau-induced neurotoxicity. These and other observations described in subsequent sections of this paper suggest that (1) In addition to its role as a cotranslational chaperone for tubulin, Pfdn5 has direct roles in the stability of mature microtubule filaments, and (2) Pfdn5 stabilizes microtubules, prevents neuronal loss, and delays the onset of Tau-induced neurotoxicity. Since the overexpression of Pfdn5 restored the Tau-induced neurological abnormalities to the control levels without causing any detectable changes in synaptic morphology, cognitive impairment, or organismal health, we suggest that Pfdn5 could be a possible therapeutic target for Tauopathies.

Results

A reverse-genetic screen of *Drosophila* chaperones identified Prefoldins as genetic modifiers of Tau-induced neurodegeneration

To identify chaperones that modulate Tau-induced neurodegeneration, we performed a screen for chaperones whose RNAi-based knockdown would modify progressive cytotoxicity observed in the eyes of *Drosophila* expressing human Tau^{V337M}. We used the hTau^{V337M} model because its expression in the eye resulted in moderate phenotypes, and therefore, allowed us to score for both enhancement or suppression of eye-ommatidial degeneration visibly (Chau et al. 2006 [\[1\]](#); Blard et al. 2007 [\[2\]](#)). We coexpressed this transgenic construct (*UAS-hTau^{V337M}*) with each of 109 RNAi lines (targeting 64 chaperones) in the *Drosophila* eye using the pan-retinal driver *GMR-Gal4* and examined how each RNAi line influenced ommatidial degeneration in hTau^{V337M}-expressing flies, seven days post-eclosion (Figure S1). We identified 20 chaperones that enhanced the neurotoxicity and 15 that suppressed hTau^{V337M}-induced ommatidial degeneration (Figure 1 [\[3\]](#) and Figure S1). Consistent with the previous studies (Petrucelli et al. 2004 [\[4\]](#); Kundel et al. 2018 [\[5\]](#)), we found that the knockdown of Hsp70 enhanced Tau-induced neurodegeneration (Figure S1I-I' and AH), validating the authenticity of this screen. In addition, the screen identified novel candidate Tau modulators. These notably included *Drosophila* orthologs of Prefoldins, tubulin binding cofactor E (TBCE), and chaperonin containing TCP1 (CCT), chaperones known to co-translationally regulate proper folding of tubulin or actin monomers (Figure 1C-L [\[6\]](#)). Knockdown of Prefoldin subunits by different independent RNAi constructs strongly enhanced the ommatidial degeneration (Figure 1C-I [\[7\]](#) and Figure N). Similar effects were also seen following the knockdown of CCT or TBCE (Figure 1J-L [\[8\]](#) and Figure N).

The effect of Pfdn5 knockdown on Tau-induced eye degeneration, measured by quantifying the percentage of the degenerated eye area, was particularly robust compared to the hTau^{V337M} expressing flies (*GMR-Gal4>UAS-hTau^{V337M}*: 42.74 ± 1.59% vs *GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5 RNAi*: 85.06 ± 3.62%; *p* < 0.001) (Figure 1M [\[9\]](#) and Figure S2). Three independent Pfdn5 RNAi lines targeting different regions of Pfdn5 transcripts enhanced the Tau-phenotypes (Supplemental Table 1). The quantitative RT-PCR (qPCR) analysis revealed a reduction in ~50-60% *pfdn5* transcript upon knockdown with Actin-Gal4 (Figure S2). Moreover, prior cell culture studies have supported the idea that Prefoldin functions to regulate the solubility of aggregate-prone proteins (Sakono et al. 2008 [\[10\]](#); Sorgjerd et al. 2013 [\[11\]](#)). For instance, the deletion of Pfdn is accompanied by the accumulation of PolyQ or Htt aggregates in cell lines (Tashiro et al. 2013 [\[12\]](#); Takano et al.

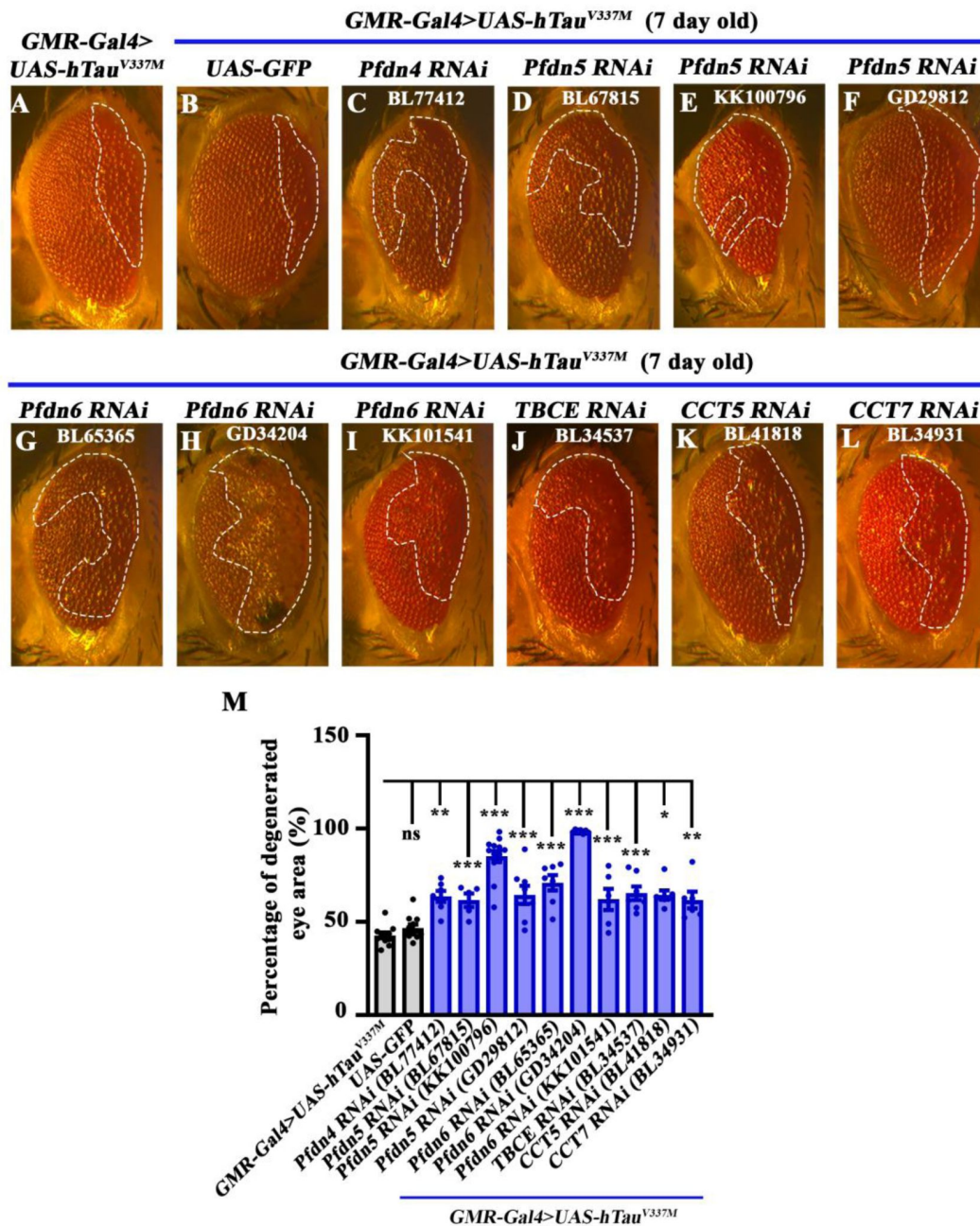


Figure 1

Prefoldin cochaperones are genetic modifiers of Tau-induced eye degeneration

(A-L) Bright-field images of 7-day old *Drosophila* eyes expressing (A) hTau^{V337M} (control), (B) GMR-Gal4>UAS-hTau^{V337M}; UAS-GFP (Gal4 dilution control), (C) GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn4 RNAi, (D), (E), (F) GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5 RNAi, (G), (H), (I) GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6 RNAi, (J) GMR-Gal4>UAS-hTau^{V337M}; UAS-TBCE RNAi, (K) GMR-Gal4>UAS-hTau^{V337M}; UAS-CCT5 RNAi, (L) GMR-Gal4>UAS-hTau^{V337M}; UAS-CCT7 RNAi. **(M)** Histogram showing the percentage of degenerated area in eyes of 7-day old flies of genotypes: UAS-hTau^{V337M}/+; GMR-Gal4/+ (42.74 ± 1.59), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-GFP (46.7 ± 1.79), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn4 RNAi (BL77412; 63.9 ± 2.95), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn5 RNAi (BL67815; 61.6 ± 3.6), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn5 RNAi (KK100796; 85.06 ± 3.01), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn5 RNAi (GD29812; 64.44 ± 4.91), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn6 RNAi (BL65365; 70.95 ± 4.17), UAS-hTau^{V337M}/+; GMR-Gal4/+; UAS-Pfdn6 RNAi/+ (GD34204; 98.57 ± 0.4), UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn6 RNAi (KK101541; 77.11 ± 2.96), UAS-hTau^{V337M}/+; GMR-Gal4/+; UAS-TBCE RNAi/+ (BL34537; 62.01 ± 5.65), UAS-hTau^{V337M}/+; GMR-Gal4/+; UAS-CCT5 RNAi/+ (BL41818; 64.36 ± 2.43), UAS-hTau^{V337M}/+; GMR-Gal4/+; UAS-CCT7 RNAi/+ (BL34931; 61.67 ± 4.47). *p<0.05; ***p<0.001; ns, not significant. At least 6 brightfield eye images of each genotype were used for quantification.

2014). In order to explore whether this proposed role of Prefoldin could be relevant to the control of Tau aggregation and neurodegeneration *in vivo*, and the finding that Pfdn5 is downregulated in Alzheimer's patients (Ji et al. 2022; Askenazi et al. 2023), we selected Pfdn5 for careful and detailed analysis.

Prefoldin 5 regulates microtubule organization and levels of tubulin monomers

Pfdn5 is a component of the hetero-hexameric Prefoldin complex, which regulates the folding of nascent actin and tubulin monomers (Vainberg et al. 1998; Leroux et al. 1999). To rigorously analyse the neuronal functions of Pfdn5, we generated loss-of-function mutants of Pfdn5 using CRISPR/Cas9-based genome editing. We created two independent *Pfdn5* mutants ($\Delta Pfdn5^{15}$ and $\Delta Pfdn5^{40}$) using two distinct pairs of gRNAs (Figure 2A). Both these *Pfdn5* mutants were null alleles, as no *Pfdn5* transcript was detected in the mutants (Figure S3A). Both homozygous and transheteroallelic mutants of Pfdn5 exhibited larval lethality at the L3 developmental stage (Figure 2A). These findings suggest that *Pfdn5* is an essential gene required ubiquitously for survival.

Analysis of third instar larval NMJ in *Pfdn5* mutants revealed the presence of several supernumerary boutons (Figure S3B-I). Prior work has shown that induction of supernumerary boutons can result from destabilizing the microtubule cytoskeleton at the NMJ (Xiong et al. 2013; Saunders et al. 2022). We therefore investigated whether loss of Pfdn5 can influence axonal microtubules. We visualized microtubules using the monoclonal antibody 22C10, which labels the microtubule-associated protein Futsch in neurons. Futsch-positive loops are seen within a subset of stable presynaptic boutons (Roos et al. 2000b). Boutons containing such loops were greatly reduced in the *Pfdn5* mutant (control: 19.98 ± 2.18 vs $\Delta Pfdn5^{15/40}$: 7.72 ± 1.62 ; $p < 0.001$), and this reduction was restored to the control levels ($\Delta Pfdn5^{15/40}$: 7.72 ± 1.62 vs *Elav-Gal4/+; UAS-Pfdn5/+; $\Delta Pfdn5^{15/40}$* : 27.39 ± 2.21 ; $p < 0.001$) upon pan-neuronal expression of a *Pfdn5* transgene in *Pfdn5* mutant background (Figure 2B-F). Consistent with this, we found a significant reduction in the intensity of acetylated tubulin, which represents long-lived, stable microtubules, at the synapses and in the muscle of *Pfdn5* mutant compared to the control (synapses - control: 0.54 ± 0.04 vs. $\Delta Pfdn5^{15/40}$: 0.36 ± 0.03 ; $p < 0.001$; muscles - control: 530.1 ± 71.56 vs. $\Delta Pfdn5^{15/40}$: 141.9 ± 8.43 ; $p < 0.001$) (Figure S4A-H). The reduced acetylated tubulin intensity at *Pfdn5* null mutant synapses was restored by the pan-neuronal expression of the *Pfdn5* transgene (Figure S4A-H). Taken together, the data support a function of Pfdn5 in regulating microtubule stability and organization *in vivo*.

Pfdn5 is also a known cotranslational chaperone for monomeric tubulin (Gestaut et al. 2022). We therefore also examined whether and how loss of Pfdn5 altered levels of tubulin monomers. Using RT-PCR, we found that the transcript level of tubulin was not altered in the *Pfdn5* mutants, suggesting that Pfdn5 does not alter tubulin gene transcription or transcript stability (Figure S4I). In contrast, western blotting revealed about 85% reduction in the α -tubulin monomers (control: 1.00 ± 0.00 vs $\Delta Pfdn5^{15/40}$: 0.11 ± 0.04 ; $p < 0.001$), about 70 percent reduction in β -tubulin (control: 1.00 ± 0.00 vs $\Delta Pfdn5^{15/40}$: 0.31 ± 0.09 $p < 0.001$) and about 60% reduction in ace-tubulin levels (control: 1.00 ± 0.00 vs $\Delta Pfdn5^{15/40}$: 0.46 ± 0.04 ; $p < 0.001$) in the *Pfdn5* mutant compared to the control (Figure 2G-J). These reductions were not seen when a *Pfdn5* transgene was ubiquitously expressed in a *Pfdn5* mutant background (Figure 2G-J). The observation that β -actin levels remained unchanged (Figure 2G) in *Pfdn5* mutants is not particularly surprising, given a prior report indicating a differential requirement of the *Drosophila* Prefoldin complex in actin and tubulin biogenesis (Delgehyr et al. 2012). Together, these data are consistent with: a) *Drosophila* Pfdn5 serving as an evolutionarily conserved function in cotranslational folding of monomeric tubulin; and b) an additional role may be performed by Pfdn5 in the regulation of mature microtubule filaments in neurons. One or both of these functions could potentially be required for the stabilization of axonal microtubules.

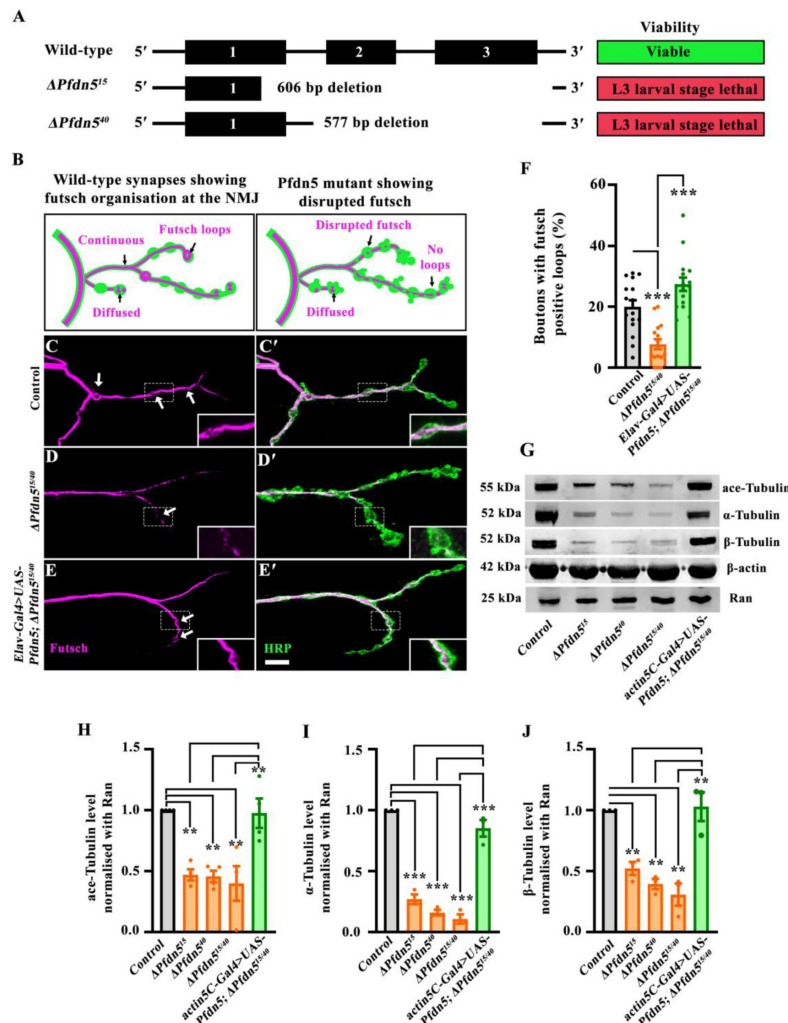


Figure 2

Loss of Pfdn5 disrupts microtubule organisation

(A) Generation of a loss-of-function mutant of Pfdn5 using CRISPR/Cas9-based genome editing. Schematic representation of the Pfdn5 genomic organization showing exons (solid black boxes, 1-3) and introns (thin black lines). Two loss-of-function Pfdn5 mutants with 606 bp (line-15) or 577 bp (line-40) deletion were obtained. Both the mutant lines are third-instar larval lethal. **(B)** Schematic representation of Futsch loop organization in muscle 4 of A2 hemisegment in wild-type or *Pfdn5* mutant. *Pfdn5* mutant shows diffused Futsch loop organization and reduced loops at the terminal boutons. **(C-E')** Confocal images of NMJ synapses at muscle 4 of A2 hemisegment showing Futsch loops in (C-C') control, (D-D') $\Delta Pfdn^{515/40}$, (E-E') *Elav-Gal4>UAS-Pfdn5; $\Delta Pfdn^{515/40}$* double immunolabeled with neuronal membrane marker, HRP (green) and 22C10 antibody against microtubule-associated protein, Futsch (magenta). The scale bar in E' for (C-E') represents 10 μ m. The inset shows a 2.0x magnified futsch loop. **(F)** Histogram showing the percentage of Futsch positive loops from muscle 4 at A2 hemi segment in control (19.98 ± 2.18), $\Delta Pfdn^{515/40}$ (7.72 ± 1.62), *Elav-Gal4>UAS-Pfdn5; $\Delta Pfdn^{515/40}$* (27.39 ± 2.21). *** $p < 0.001$. At least 16 NMJs of each genotype were used for quantification. **(G)** Western blot showing protein levels of α -Tubulin, β -Tubulin, ace-Tubulin, and Actin in control, $\Delta Pfdn^{515/40}$, $\Delta Pfdn^{540/40}$, $\Delta Pfdn^{515/40}$, and *actin5C-Gal4>UAS-Pfdn5; $\Delta Pfdn^{515/40}$* . Ran protein levels were used as an internal loading control. **(H)** Histogram showing the percentage of ace-Tubulin normalized with Ran in control (1.00 ± 0.00), $\Delta Pfdn^{515/40}$ (0.50 ± 0.05), $\Delta Pfdn^{40/40}$ (0.49 ± 0.04), $\Delta Pfdn^{515/40}$ (0.46 ± 0.04), *actin5C-Gal4/UAS-Pfdn5; $\Delta Pfdn^{515/40}$* (1.05 ± 0.13). *** $p < 0.01$. Three independent western blots were used for quantification. **(I)** Histogram showing percentage of α -Tubulin normalised with Ran in control (1.00 ± 0.00), $\Delta Pfdn^{515/40}$ (0.27 ± 0.04), $\Delta Pfdn^{40/40}$ (0.16 ± 0.02), $\Delta Pfdn^{515/40}$ (0.11 ± 0.04), *actin5C-Gal4/UAS-Pfdn5; $\Delta Pfdn^{515/40}$* (0.85 ± 0.07). *** $p < 0.001$. Three independent western blots were used for quantification. **(J)** Histogram showing percentage of β -Tubulin normalised with Ran in control (1.00 ± 0.00), $\Delta Pfdn^{515/40}$ (0.52 ± 0.04), $\Delta Pfdn^{40/40}$ (0.39 ± 0.04), $\Delta Pfdn^{515/40}$ (0.30 ± 0.09), *actin5C-Gal4/UAS-Pfdn5; $\Delta Pfdn^{515/40}$* (1.03 ± 0.11). *** $p < 0.01$. Three independent western blots were used for quantification.

Pfdn5 is a novel neuronal microtubule-associated protein

To assess the function of Pfdn5, we began by generating antisera against the full-length Pfdn5. Western blot analysis using Pfdn5 antibody revealed a protein band of ~18 kDa in the larval lysates that was absent in *Pfdn5* mutants, thus validating the specificity of the antibody (Figure S5A). In larval fillets examined by immunocytochemistry, anti-Pfdn5 staining was dramatically reduced in the mutant, with faint residual staining consistent with a low level of maternally provided Pfdn5 (Figure 3A-B''' and Figure S5B-E). The Pfdn5 staining was restored upon expression of the Pfdn5 transgene in the Pfdn5 mutant background (Figure 3B-C''' and Figure S5D-E). However, heterozygous mutants of Pfdn5 ($\Delta Pfdn5^{15/+}$ and $\Delta Pfdn5^{40/+}$) revealed no significant reduction in the levels of Pfdn5 or tubulin (Figure S5F-H). Careful examination shows that Pfdn5 colocalized with axonal microtubule (labelled with α -tubulin antibody) in wild-type larvae (Figure 3A-A'''). The level of α -tubulin was significantly reduced in the absence of Pfdn5 (Figure 3A-C'''). The Pearson's correlation coefficient of 0.60 ± 0.02 across pixels labelled by α -tubulin and Pfdn5 in axons further strengthens that a tight colocalization exists between Pfdn5 and neuronal microtubules (Figure 3D).

To further test whether Pfdn5 associates with microtubules, we stabilized microtubules using Taxol and performed a microtubule-binding experiment (Ando et al. 2016). We found Pfdn5 in the pellet fraction when microtubules were stabilized with Taxol, but not under the condition where the microtubules were severed using Nocodazole. The quantification revealed a substantially higher Pfdn5 binding to stabilized microtubules when compared to non-stabilized microtubule control (Taxol: 36.67 ± 7.56 , vs control: 7.83 ± 2.92 ; $p < 0.001$) (Figure 3E-G). Consistent with our immunocytochemistry results, we found that Pfdn5 binds with the Taxol-stabilised microtubule. This unexpected localization of Pfdn5 to neuronal microtubule filaments and its binding to the stable microtubule points to a role for this protein beyond its function as a cotranslational chaperone, potentially in the organization or the stability of axonal microtubule cytoskeleton.

Loss of Pfdn5 phenocopies and synergistically aggravates the Tau-induced synaptic defects

Since microtubules regulate morphological features of synapses (Roos et al. 2000b) and loss of Pfdn5 resulted in reduced stable microtubules, we next asked if *Pfdn5* mutants show distinctly altered synaptic architecture at their NMJ. Both the homozygous and heteroallelic mutant combination showed numerous supernumerary boutons with altered synaptic morphology when compared to the control (control: 2.25 ± 0.41 vs. $\Delta Pfdn5^{15/40}$: 18.25 ± 1.27 ; $p < 0.001$) (Figure 4A-B and 4J). Increased supernumerary boutons in the *Pfdn5* mutant were completely restored upon pan-neuronal expression of a wild-type *Pfdn5* transgene using *Elav-Gal4* in the *Pfdn5* mutant background ($\Delta Pfdn5^{15/40}$: 18.25 ± 1.27 vs *Elav-Gal4*+/+; *UAS-Pfdn5*+/+; $\Delta Pfdn5^{15/40}$: 2.94 ± 0.67 ; $p < 0.001$) (Figure 4C and 4J) further confirming that these phenotypes were caused by *Pfdn5* mutations and not unknown potential background mutations. Expression of *Pfdn5* in muscles using *mef2-Gal4* failed to rescue the lethality or the synaptic defects ($\Delta Pfdn5^{15/40}$: 18.25 ± 1.27 vs *UAS-Pfdn5*+/+; *mef2-Gal4*, $\Delta Pfdn5^{15/40}$: 15.60 ± 0.86 ; $p > 0.75$) (Figure 4D and 4J), suggesting that the synaptic phenotype arises due to loss of Pfdn5 in neurons and not muscles.

Ectopic satellite boutons and disrupted microtubules that we observed in *Pfdn5* mutant larval NMJs appeared very similar to those previously described in *Drosophila* expressing hTau^{V337M} in motor neurons (Blard et al. 2007 ; Xiong et al. 2013 ; Mao et al. 2017). This apparent similarity, together with our identification of Pfdn5 as a genetic modifier of hTau^{V337M}-induced cytotoxicity, led us to more closely examine phenotypic similarities between *Pfdn5* mutant and hTau^{V337M} expressing animals, as well as genetic interactions between *Pfdn5* and hTau^{V337M}. We

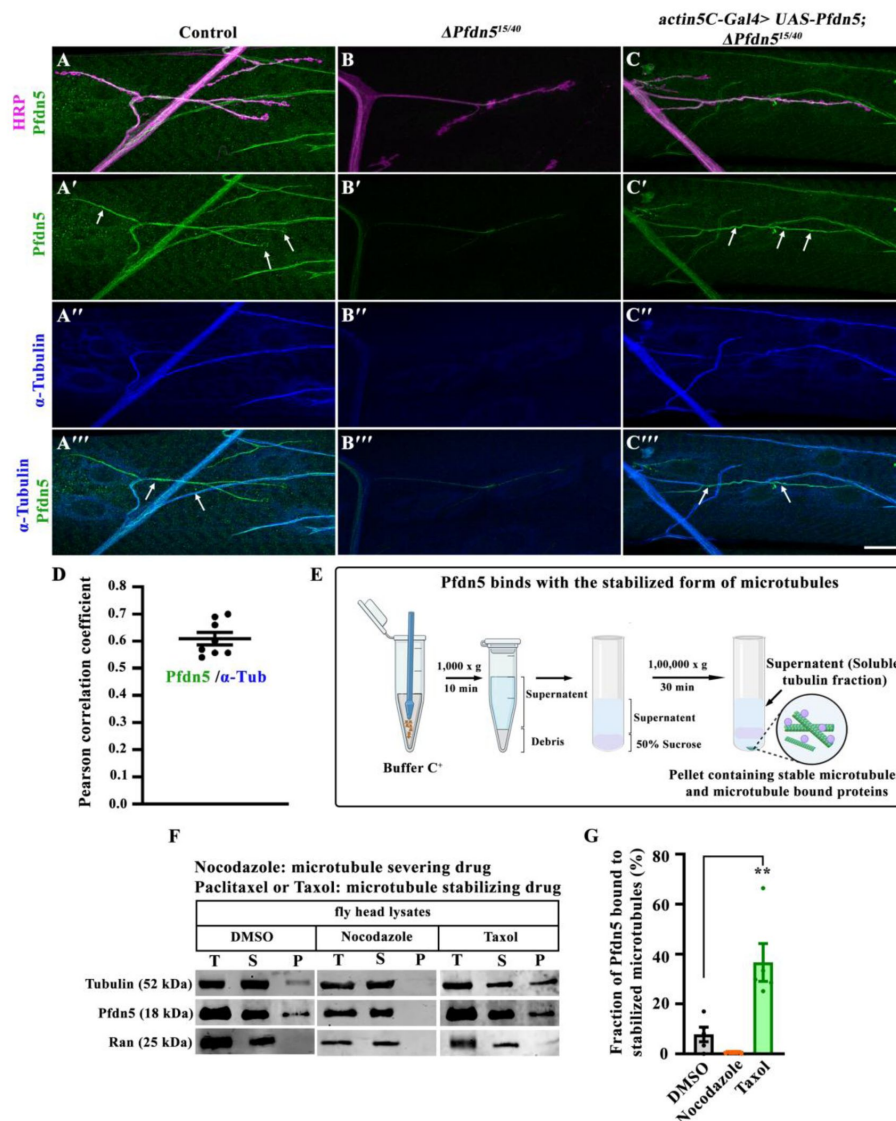


Figure 3

Pfdn5 is a novel microtubule-binding protein

(A-C''') Confocal images of NMJ synapses at muscle 4 of A2 hemisegment triple-labeled for Pfdn5 (green), α -Tubulin (blue), and HRP (magenta) showing that Pfdn5 colocalizes with microtubule cytoskeleton in (A-A''') in the wild-type larval neuronal axons and in the tracheal tubes, (B-B''') *Pfdn5* mutants show dramatically reduced Pfdn5 and α -Tubulin levels, (C-C''') *actin5C-Gal4*-mediated rescue (*actin5C-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$*) significantly restored the level of Pfdn5 and α -Tubulin. Arrows represent Pfdn5 colocalization with microtubule loops, which are not detectable in the *Pfdn5* mutants. The scale bar in C''' (for A-C''') represents 10 μ m. (D) Pearson's correlation coefficient to quantify colocalization between Pfdn5 and axonal microtubule. (E) Schematic representation of the microtubule-binding protocol. Head lysate from wild-type flies, treated with Taxol or Nocodazole, was subjected to ultracentrifugation. The 'soluble fraction' contains free tubulin whereas the 'insoluble pellet fraction' contains the stabilized microtubule along with microtubule-binding proteins, which can be detected by western blotting. The details of the protocol is described in the material and methods section. (F) Microtubule binding assay with *Drosophila* head lysate in the presence of Taxol or Nocodazole (diluted in DMSO). T represents (Total fraction: input fraction), S represents (Supernatant: free tubulin), and P represents (Pellet fraction: stabilized microtubule). Immunoblot with antibodies against α -Tubulin or Pfdn5 detected increased Pfdn5 in the pellet fraction in the presence of Taxol but not Nocodazole. The binding of Pfdn5 with stabilized microtubules was calculated as the percentage of Pfdn5 in the pellet fraction. Ran was used as the loading control. (G) Histogram showing the percentage of Pfdn5 in the pellet fraction of *in vivo* microtubule binding assay in the presence of DMSO (7.83 $\pm$ 2.92), Nocodazole (0.42 $\pm$ 0.1), or Taxol (36.67 $\pm$ 7.56). Five independent western blots were used for quantification.

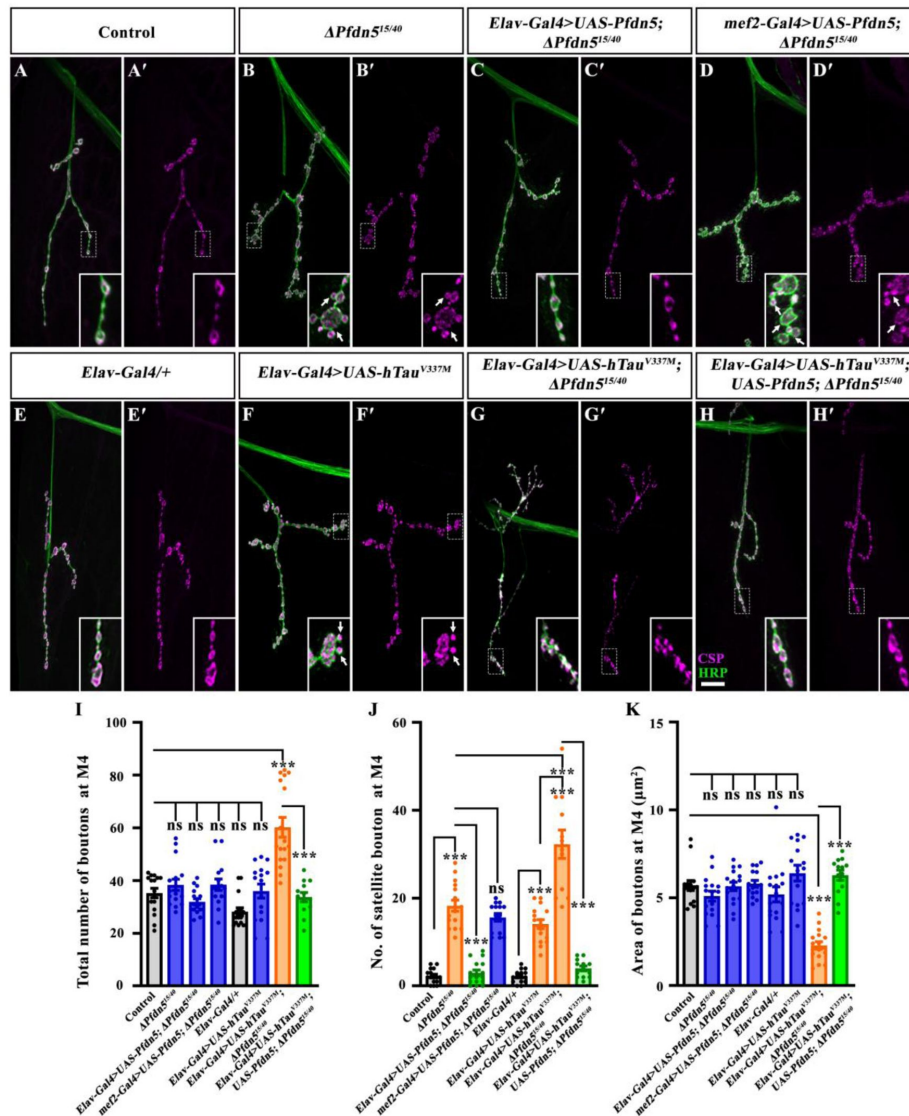


Figure 4

Loss of Pfdn5 mimics and enhances Tau-induced synaptic defects

(A-H') Confocal images of NMJ synapses at muscle 4 of A2 hemisegment showing synaptic morphology in (A-A') control, (B-B') $\Delta Pfdn5^{15/40}$, (C-C') *Elav-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* , (D-D') *mef2-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* , (E-E') *Elav-Gal4/+* (Gal4 control), (F-F') *Elav-Gal4>UAS-hTau^{V337M}*, (G-G') *Elav-Gal4>hTau^{V337M}; $\Delta Pfdn5^{15/40}$* , (H-H') *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* double immunolabeled for HRP (green), and CSP (magenta). The scale bar in H for (A-H') represents 10 μm . Arrows point to clustered satellite boutons. (I) Histogram showing total number of boutons from muscle 4 at A2 hemisegment in control (35.25 ± 1.8), $\Delta Pfdn5^{15/40}$ (38.38 ± 2.15), *Elav-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (31.94 ± 1.18), *mef2-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (38.40 ± 2.17), *Elav-Gal4/+* (28.13 ± 1.51), *Elav-Gal4>UAS-hTau^{V337M}* (36.00 ± 2.65), *Elav-Gal4>hTau^{V337M}; $\Delta Pfdn5^{15/40}$* (60.34 ± 3.76), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (33.69 ± 1.76). *** $p < 0.001$; ns, not significant. At least 12 NMJs of each genotype were used for quantification. (J) Histogram showing number of satellite boutons from muscle 4 at A2 hemisegment in control (2.25 ± 0.41), $\Delta Pfdn5^{15/40}$ (18.25 ± 1.28), *Elav-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (2.94 ± 0.67), *mef2-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (15.6 ± 0.86), *Elav-Gal4/+* (2.2 ± 0.38), *Elav-Gal4>UAS-hTau^{V337M}* (14.06 ± 1.00), *Elav-Gal4>hTau^{V337M}; $\Delta Pfdn5^{15/40}$* (32.25 ± 3.2), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (4.0 ± 0.5). *** $p < 0.001$; ns, not significant. At least 12 NMJs of each genotype were used for quantification. (K) Histogram showing bouton area from muscle 4 at A2 hemisegment in control (5.7 ± 0.29), $\Delta Pfdn5^{15/40}$ (5.1 ± 0.28), *Elav-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (5.6 ± 0.2), *mef2-Gal4>UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (5.8 ± 0.2), *Elav-Gal4/+* (5.2 ± 0.4), *Elav-Gal4>UAS-hTau^{V337M}* (6.4 ± 0.4), *Elav-Gal4>hTau^{V337M}; $\Delta Pfdn5^{15/40}$* (2.3 ± 0.2), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5; $\Delta Pfdn5^{15/40}$* (6.3 ± 0.3). *** $p < 0.001$; ns, not significant. At least 12 NMJ of each genotype were used for quantification.

first confirmed that the loss of Pfdn5 phenocopies the Tau^{V337M}-induced morphological defects at synapses (**Figure 4E-F**). We then tested the effects of loss and gain of Pfdn5 on Tau^{V337M} phenotypes.

Morphological NMJ phenotypes induced by expressing hTau^{V337M} pan-neuronally using *Elav-Gal4* were strongly enhanced in a *Pfdn5* mutant background (**Figure 4G** and **4I-K**). While satellite bouton numbers in *Elav-Gal4/UAS-hTau^{V337M}* and $\Delta Pfdn5^{15/40}$ were: 18.25 ± 1.27 and 14.06 ± 1.00 respectively, this was significantly increased in *Elav-Gal4/UAS-hTau^{V337M}; $\Delta Pfdn5^{15/40}$* combination (32.25 ± 3.22 ; $p < 0.001$) (**Figure 4J**). Subsequently, total bouton number was significantly increased in *Elav-Gal4/UAS-hTau^{V337M}; $\Delta Pfdn5^{15/40}$* combination animals (60.24 ± 3.76 ; $p < 0.001$) compared to either *Elav-Gal4/UAS-hTau^{V337M}* (36.00 ± 2.65) or $\Delta Pfdn5^{15/40}$ (38.38 ± 2.15) alone. Similarly, bouton area (in μm^2) at NMJs of *Elav-Gal4/UAS-hTau^{V337M}; $\Delta Pfdn5^{15/40}$* combination animals (2.29 ± 0.20) were substantially smaller than either *Elav-Gal4/UAS-hTau^{V337M}* (6.40 ± 0.45) or $\Delta Pfdn5^{15/40}$ (5.09 ± 0.28) alone (**Figure 4K**). Consistently, expressing a more severe form of pathological Tau (hTau^{R406W}) in the *Pfdn5* mutation background also resulted in further enhancement in the synaptic phenotypes (**Figure S6A-E**). Thus, loss of Pfdn5 aggravates not only Tau-induced eye degeneration (**Figure 1**) but also specific Tau-induced synaptic defects, suggestive of function in a common pathway. Next, we investigated whether this enhanced synaptic phenotype results from increased microtubule disruption. We found that Futsch staining was significantly reduced at the synapses when hTau^{V337M} was expressed in the *Pfdn5* mutant background (**Figure S6F-J**). These findings indicate that microtubule disruption caused by the loss of Pfdn5 contributes to the exacerbation of hTau^{V337M}-induced synaptic defects.

Loss of Pfdn5 enhances pathological Tau aggregation in larval brain and axons

Tau-induced neurotoxicity directly correlates with the extent of Tau phosphorylation and its deposition as insoluble aggregates (Chau et al. 2006; Aquino Nunez et al. 2022). Hence, we performed additional experiments to explore mechanisms by which Pfdn5 levels could influence Tau function. We considered a model in which Pfdn5 acts to prevent Tau aggregation, thereby suppressing the Tau-induced neurodegeneration. The immunocytochemistry revealed that animals lacking Pfdn5 showed a remarkable increase in Tau-aggregates in the larval brain (Tau punctae with size $> 3 \mu\text{m}^2$ per brain lobe: *Elav-Gal4/UAS-hTau^{V337M}*: 1.13 ± 0.39 vs *Elav-Gal4/UAS-hTau^{V337M}; $\Delta Pfdn5^{15/40}$* : 10.50 ± 2.57 ; $p < 0.001$) (**Figure 5A-F**). Consistent with these observations, hTau distribution in *Pfdn5* mutant axons revealed a substantially higher number of hTau-aggregates compared to animals with normal levels of Pfdn5 (*Elav-Gal4/UAS-hTau^{V337M}*: 0.25 ± 0.09 / $100 \mu\text{m}^2$ vs *Elav-Gal4/UAS-hTau^{V337M}; $\Delta Pfdn5^{15/40}$* : 2.9 ± 0.41 / $100 \mu\text{m}^2$; $p < 0.001$) (**Figure 5G-J'** and **4K**). The increased hTau puncta in *Pfdn5* mutants were significantly suppressed upon normalizing the level of Pfdn5 in neurons. Analysis of fluorescence intensity profiles across the Tau puncta showed a 4-fold increase in Tau intensity, further supporting that Tau indeed forms aggregates in the absence of Pfdn5 (**Figure S7A-C**). Additional experiments involving quantification of axonal hTau using immunofluorescence revealed that levels of Tau were significantly reduced in animals overexpressing hTau^{V337M} in *Pfdn5* mutants compared to animals expressing hTau^{V337M} alone (**Figure 5G-J''** and **L**). Similar results were obtained when the phospho-Tau antibody was used to assess Tau levels and aggregates (**Figure S7D-F**). To determine if loss of Pfdn5 specifically modulates FTDP-17-associated Tau mutations or also influences sporadic Tauopathy linked to wild-type Tau, we expressed hTau^{WT} in *Pfdn5* mutants. Interestingly, expression of wild-type Tau (hTau^{WT}) in the *Pfdn5* mutants showed an aggregation pattern in the larval optic lobes similar to that observed with Tau^{V337M} (**Figure S7G-I**). These data reveal that Pfdn5 suppresses Tau aggregation in both the brain and axons and that loss of Pfdn5 can induce the onset of multiple forms of Tauopathies.

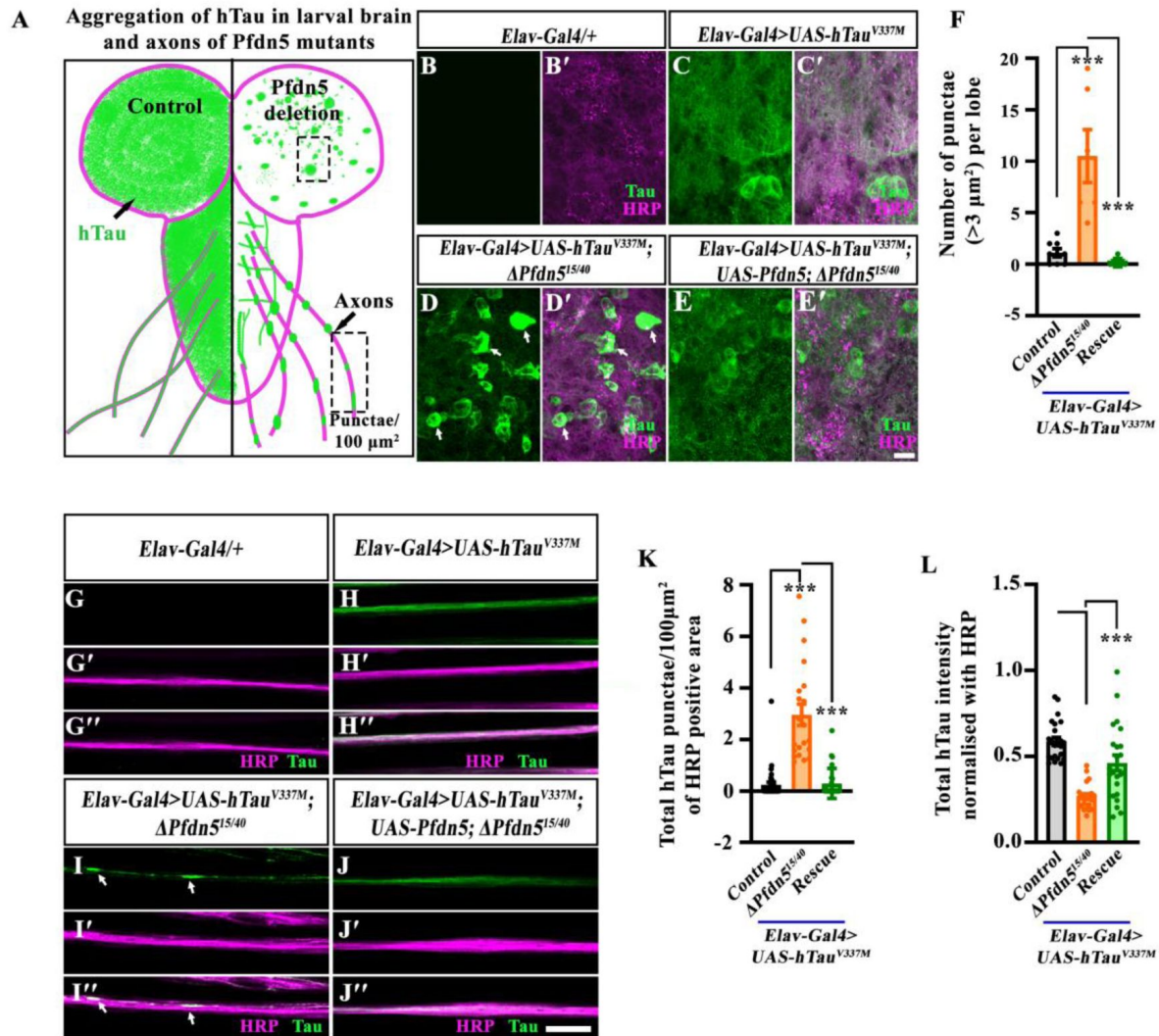


Figure 5

Loss of *Pf_{dn}5* induces formation of hTau^{V337M} aggregates in larval neurons

(A) Schematic representation of pathological hTau distribution in larval brain lobes and axons of the control (left half) or *Pf_{dn}5* mutant (right half) animals. **(B-E')** Confocal single section images of third instar larval brain in **(B-B')** *Elav-Gal4/+* (control), **(C-C')** *Elav-Gal4>UAS-hTau^{V337M}*, **(D-D')** *Elav-Gal4>UAS-hTau^{V337M}; Δ*Pf_{dn}5*^{15/40}*, **(E-E')** *Elav-Gal4>UAS-hTau^{V337M}; UAS-*Pf_{dn}5*; Δ*Pf_{dn}5*^{15/40}* double immunolabeled with neuronal membrane marker, HRP (magenta), and T46 antibody against hTau (green). The scale bar in **E'** for **(B-E')** represents 10 μm. Arrows in **D** and **D'** point to the hTau punctae/flame shaped aggregates in the brain. **(F)** Histogram showing the quantification of the number of hTau punctae (>3 μm²) in *Elav-Gal4>UAS-hTau^{V337M}* (1.13 ± 0.39), *Elav-Gal4>UAS-hTau^{V337M}; Δ*Pf_{dn}5*^{15/40}* (10.5 ± 2.57), and *Elav-Gal4>UAS-hTau^{V337M}; UAS-*Pf_{dn}5*; Δ*Pf_{dn}5*^{15/40}* (0.17 ± 0.17). ****p*<0.001. At least 6 optic lobes of each genotype were used for quantification. **(G-J'')** Confocal single section images of third instar larval axons in **(G-G'')** *Elav-Gal4/+* (control), **(H-H'')** *Elav-Gal4>UAS-hTau^{V337M}*, **(I-I'')** *Elav-Gal4>UAS-hTau^{V337M}; Δ*Pf_{dn}5*^{15/40}*, **(J-J'')** *Elav-Gal4>UAS-hTau^{V337M}; UAS-*Pf_{dn}5*; Δ*Pf_{dn}5*^{15/40}* double immunolabeled for HRP (magenta), and T46 antibody against hTau (green). The scale bar in **J''** for **(G-J'')** represents 10 μm. Arrows in **I** and **I''** point to the hTau^{V337M} aggregates in axons. **(K)** Histogram showing the quantification of the number of hTau punctae normalized with HRP positive area in *Elav-Gal4>UAS-hTau^{V337M}* (0.25 ± 0.1), *Elav-Gal4>UAS-hTau^{V337M}; Δ*Pf_{dn}5*^{15/40}* (2.96 ± 0.4), and *Elav-Gal4>UAS-hTau^{V337M}; UAS-*Pf_{dn}5*; Δ*Pf_{dn}5*^{15/40}* (0.29 ± 0.1). ****p*<0.001. At least 20 axons from 8 animals of each genotype were used for quantification. **(L)** Histogram showing the intensity of total hTau normalized with HRP in *Elav-Gal4>UAS-hTau^{V337M}* (0.59 ± 0.02), *Elav-Gal4>UAS-hTau^{V337M}; Δ*Pf_{dn}5*^{15/40}* (0.27 ± 0.02), and *Elav-Gal4>UAS-hTau^{V337M}; UAS-*Pf_{dn}5*; Δ*Pf_{dn}5*^{15/40}* (0.46 ± 0.4). ****p*<0.001. At least 20 axons from 8 animals of each genotype were used for quantification.

Next, in order to assess the nature of Tau aggregates, we performed a Tau solubility assay (Vourkou et al. 2023 [DOI](#)). We found that in the absence of Pfdn5, hTau levels were reduced in the supernatant fraction and increased in the pellet fraction (Figure S8A-C). Previous *in vitro* studies have shown that hTau can undergo liquid-liquid phase separation to form Tau droplets, which can be dissolved by 1,6-Hexanediol; in contrast, stable aggregates of Tau remain intact in the presence of 1,6-Hexanediol (Wegmann et al. 2018 [DOI](#)). To determine whether the Tau puncta formed in the absence of Pfdn5 represented stable aggregates or phase-separated condensates, we treated the larval fillets with increasing concentrations of 1,6-Hexanediol. We observed that the Tau puncta formed in *Pfdn5* mutants remain intact even at 5% 1,6-Hexanediol (Figure S8D-L), indicating that loss of Pfdn5 promotes the formation of stable Tau aggregates.

Colocalization analysis with the HRP and Elav showed that these aggregates are mainly present in the cell body and axons (Figure S8D-N'). Tau aggregation is known to occur either through Tau hyperphosphorylation or disruption of microtubules (Jackson et al. 2002 [DOI](#); Okenve-Ramos et al. 2024 [DOI](#)). To examine the role of phosphorylation, we stained the aggregates with the phospho-Tau-specific antibody, AT8, which detects pSer202 and pThr205 (Goedert et al. 1995 [DOI](#)). The AT8 antibody failed to label all aggregates detected by the total Tau antibody D5D8N, suggesting that Tau aggregates formed in the loss of Pfdn5 animals might have conformational heterogeneity (Figure S9A-E). Next, we inhibited Tau phosphorylation using 20 mM LiCl (Cowan et al. 2010 [DOI](#)) in *Elav-Gal4>UAS-hTau^{V337M}, ΔPfdn5^{15/40}* larvae. We found that LiCl treatment did not significantly alter aggregate formation, indicating a minimal role of Tau hyperphosphorylation in the formation of aggregates in the absence of Pfdn5 (Figure S9F-J). Together, (a) the marked increase of aggregated hTau in the absence of Pfdn5 and (b) enhancement of the Tau-associated phenotypes by loss of Pfdn5 indicate that Pfdn5 prevents the transition of hTau from soluble and/or microtubule-associated state to an aggregated, insoluble, and pathogenic state in a Tau-hyperphosphorylation independent manner.

Neuronal overexpression of Pfdn5 or Pfdn6 ameliorates the hTau-induced age-dependent progression of the neurodegeneration

The observations above indicate that loss of Pfdn5 enhances the neurotoxicity in the *Drosophila* Tauopathy model. We therefore further tested whether overexpression of Pfdn5 could alleviate Tau-induced developmental toxicity in the ommatidia. We examined the effects of Pfdn5/6 overexpression on hTau^{V337M}-induced ommatidial degeneration. *GMR-Gal4*-mediated overexpression of Pfdn5 or Pfdn6 in eyes significantly rescued hTau^{V337M}-induced ommatidial degeneration in flies (Figure 6A-D' [DOI](#) and Figure S10). We found that *UAS-hTau^{V337M}/+; GMR-Gal4/+* flies showed $29.12 \pm 2.3\%$ fused ommatidia and $70.96 \pm 3.00\%$ degenerated eye area. This ommatidial degeneration was greatly suppressed when either Pfdn5 (% fused ommatidia, *UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn5*: 2.98 ± 0.31 ; $p < 0.001$; % degenerated eye area, *UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn5*: 5.30 ± 0.94 ; $p < 0.001$) or Pfdn6 (% fused ommatidia, *UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn6*: 2.78 ± 0.60 ; $p < 0.001$; % degenerated eye area *UAS-hTau^{V337M}/+; GMR-Gal4/UAS-Pfdn6*: 3.83 ± 1.37 ; $p < 0.001$) was coexpressed with pathological hTau (Figure 6A-F [DOI](#)). In order to ascertain that the suppression of ommatidial degeneration was not due to the Gal4 dilution, we coexpressed the neutral gene product GFP along with hTau^{V337M}. As expected, we found no change in the Tau-induced eye phenotype when expressed alone or with GFP (Figure S11A-B). These data suggest that the suppression of Tau-induced neurotoxicity was due to the expression of Pfdn5 or Pfdn6.

Overexpression of Pfdn5 suppressed not only hTau^{V337M}-induced neurotoxicity but also in a different Tauopathy model, hTau^{R406W}, which causes more severe neurotoxicity than hTau^{V337M} in the fly compound eye (degenerated eye area: *GMR-Gal4>UAS-hTau^{R406W}*: 82.15 ± 3.194 vs. *GMR-Gal4>UAS-hTau^{R406W}; UAS-Pfdn5*: 63.11 ± 3.49) (Figure S11C-E), indicating that Pfdn5 can mitigate

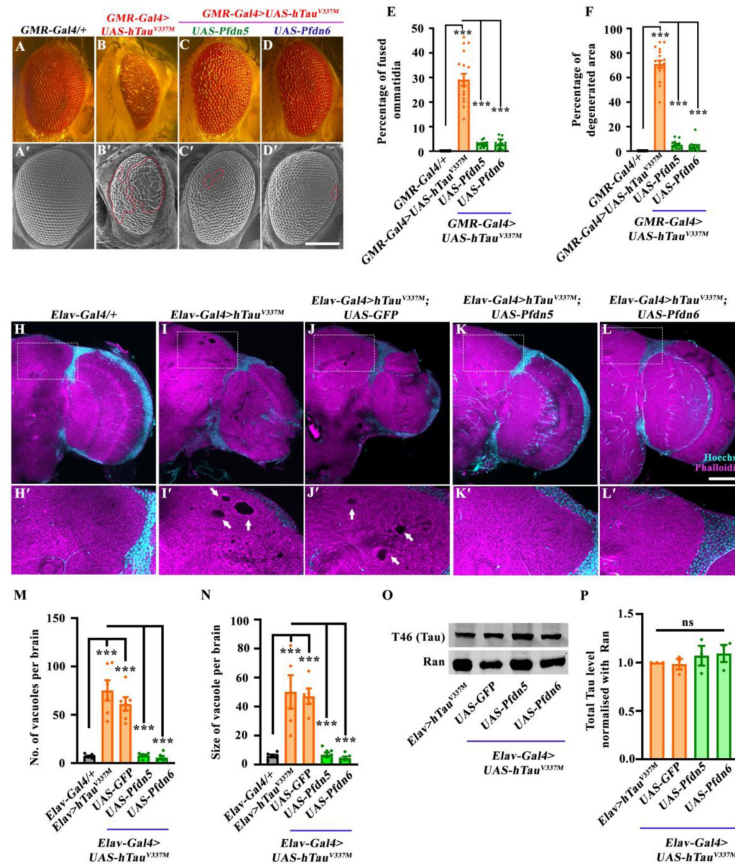


Figure 6

Over-expression of Pfdn5 or Pfdn6 suppresses age-dependent progression of hTau-induced neurodegeneration

(A-D) Bright-field images of *Drosophila* eyes expressing (A) *GMR-Gal4/+* (control), (B) *GMR-Gal4>UAS-hTau^{V337M}*, (C) *GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5*, (D) *GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6*. **(A'-D')** Scanning electron microscopic images of *Drosophila* eyes expressing (A') *GMR-Gal4/+* (control), (B') *GMR-Gal4>UAS-hTau^{V337M}*, (C') *GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5*, (D') *GMR-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6*. **(E)** Histogram showing the percentage of fused ommatidia in *GMR-Gal4/+* (0.00 ± 0.00), *UAS-hTau^{V337M}/+* (29.12 ± 2.37), *GMR-Gal4>UAS-hTau^{V337M}/+; UAS-Pfdn5* (2.98 ± 0.31), *UAS-hTau^{V337M}/+; GMR-Gal4>UAS-Pfdn6* (2.78 ± 0.60). $***p < 0.001$. At least 12 SEM eye images of each genotype were used for quantification. **(F)** Histogram showing percentage of degenerated area in *GMR-Gal4/+* (0.00 ± 0.00), *UAS-hTau^{V337M}/+* (70.96 ± 3.00), *GMR-Gal4>UAS-hTau^{V337M}/+; UAS-Pfdn5* (5.30 ± 0.94), *UAS-hTau^{V337M}/+; GMR-Gal4>UAS-Pfdn6* (3.83 ± 1.37). $***p < 0.001$. At least 12 SEM eye images of each genotype were used for quantification. **(H-L)** Confocal images of a single section of 30-day old adult brain in (H) *Elav-Gal4/+* (control), (I) *Elav-Gal4>UAS-hTau^{V337M}*, (J) *Elav-Gal4>hTau^{V337M}; UAS-GFP*, (K) *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5*, (L) *Elav-Gal4>hTau^{V337M}; UAS-Pfdn6* double immunolabeled with Hoechst (cyan), and Phalloidin (magenta). The insets represent the 3x magnified portion of the image. Arrows point to the pathological vacuolar structures. The scale bar in L for (H-L) represents 20 μm. **(M)** Histogram showing the quantification of number of vacuoles in 30-day old adult brain in *Elav-Gal4/+* (7.13 ± 0.58), *Elav-Gal4>UAS-hTau^{V337M}* (75.17 ± 10.47), *Elav-Gal4>hTau^{V337M}; UAS-GFP/+* (61.17 ± 6.91), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5/+* (7.86 ± 0.91), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn6/+* (5.67 ± 1.69). $***p < 0.001$. At least 6 brains of each genotype were used for quantification. **(N)** Histogram showing the quantification of vacuole size (in μm²) in 30-day old adult brain in *Elav-Gal4/+* (5.9 ± 0.92), *Elav-Gal4>UAS-hTau^{V337M}* (50.06 ± 11.52), *Elav-Gal4>hTau^{V337M}; UAS-GFP/+* (47.17 ± 5.39), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5/+* (6.76 ± 1.57), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn6/+* (4.36 ± 1.08). $***p < 0.001$. At least 6 brains of each genotype were used for quantification. **(O)** Western blot showing protein levels of total Tau in *Elav-Gal4>UAS-hTau^{V337M}*, *Elav-Gal4>hTau^{V337M}; UAS-GFP* (Gal4-dilution control), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5*, *Elav-Gal4>hTau^{V337M}; UAS-Pfdn6*. Ran protein levels were used as an internal loading control. **(P)** Histogram showing the percentage of total Tau (T46) normalized with Ran in *Elav-Gal4>UAS-hTau^{V337M}* (1.00 ± 0.00), *Elav-Gal4>hTau^{V337M}; UAS-GFP* (0.98 ± 0.05), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn5* (1.07 ± 0.10), *Elav-Gal4>hTau^{V337M}; UAS-Pfdn6* (1.09 ± 0.09). ns, not significant. Three independent western blots were used for quantification.

the neurodegeneration caused by at least one another variant/structural conformations of hTau. Moreover, neuronal expression of Pfdn5 and Tau^{WT} or Tau^{V337M} rescues the synaptic defects (Figure S11F-L), suggesting that Pfdn5 can alleviate multiple forms of Tauopathy.

A key pathological feature of Tau-induced neurodegeneration is age-dependent brain vacuolization in *Drosophila* and brain atrophy in humans, both of which are neuropathological hallmarks directly indicative of neuronal loss. Brain vacuolization is observed in *Drosophila* models of Tauopathy (Byrns et al. 2021 [\[1\]](#)). We therefore tested whether elevating the expression level of Pfdn5 or Pfdn6 could mitigate the appearance of vacuoles that can be detected in the brains of 30-day-old flies expressing hTau (Wittmann et al. 2001 [\[2\]](#)). We neuronally coexpressed hTau^{V337M} alone or together with Pfdn5 or Pfdn6 using *Elav-Gal4* and examined whole-mount brain preparations stained with rhodamine-phalloidin using confocal microscopy (Behnke et al. 2021 [\[3\]](#)). Consistent with the previous reports, we found several large vacuoles in the 30-day flies expressing hTau^{V337M} compared to the 2-day or 14-day-old brains (Figure 6H-J [\[4\]](#) and Figure S12). The average number and size of the vacuoles in *Elav-Gal4>UAS-hTau^{V337M}* (75.17 ± 10.47 vacuoles/brain, and 50.06 ± 11.52 µm average vacuole size) were far higher compared to the control *Elav-Gal4/+* (7.13 ± 0.58 vacuoles/brain, and 5.9 ± 0.92 µm average vacuole size) (Figure 6M-N [\[5\]](#)). Coexpression of Pfdn5 or Pfdn6 with hTau^{V337M} significantly reduced the number of vacuoles in *Elav-Gal4>UAS-hTau^{V337M}* flies. Indeed, vacuole numbers and size were restored to near control levels in *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5/+* (7.86 ± 0.91 and 6.76 ± 1.57 µm) and *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6/+* flies (5.67 ± 1.69 and 4.36 ± 1.08 µm), respectively (Figure 6K-N [\[6\]](#)). Next, we asked whether the rescue of Tau phenotypes by Pfdn5 was due to enhanced degradation of the Tau protein in the brain. Western blot analysis revealed no significant change in the Tau levels in Pfdn5 or Pfdn6 rescue animals when compared to those expressing Tau alone (Figure 6O-P [\[7\]](#)), suggesting that Pfdn5 mitigates Tau aggregates by stabilizing the microtubules rather than protein degradation. Altogether, these data indicate that increased expression of Pfdn5 or Pfdn6 can remarkably counteract neuronal loss and delay the onset and progression of the neurodegenerative cascade induced in Tauopathy through a mechanism that involves Pfdn-mediated microtubule stabilization.

Expression of Pfdn5 or Pfdn6 suppresses

Tau-induced memory impairment

Cognitive decline is a common preclinical and early feature of Tauopathies (Hanseeuw et al. 2019 [\[8\]](#)). Hence, we further examined whether Pfdn5 or Pfdn6 overexpression could rescue cognitive, memory, and behavioural deficits caused by hTau^{V337M} in the *Drosophila* brain (Orr et al., 2017). To test the memory impairments, we used a recently developed method to assess long-term aversive olfactory conditioning memory (Mohandasan et al. 2022 [\[9\]](#)). In this method, flies learn to associate bitter food (CuSO₄) with an odor (2,3 butanedione (2,3 BD)) over 8 training cycles. Memory is assessed 24 hours later as avoidance of 2,3-BD in a Y-maze (Figure 7A [\[10\]](#)). Untrained flies responded normally to the odor, while trained flies avoided it, indicating proper memory performance towards the conditioned odorant. Control flies (*UAS-hTau^{V337M}/+*) showed normal memory (*UAS-hTau^{V337M}/+*: naïve, 23.66 ± 2.42 v/s trained, 10.4 ± 2.14; p < 0.001) (Figure 7B [\[11\]](#)). However, animals expressing hTau^{V337M} in neurons (*Elav-Gal4>UAS-hTau^{V337M}*: naïve, 28.39 ± 3.47 v/s trained, 29.13 ± 4.65; p = 0.90) showed no difference between naïve and trained genotype (Figure 7C [\[12\]](#)).

Next, we assessed whether the expression of Prefoldins impacted the memory deficit phenotype of flies expressing hTau^{V337M}. We first examined the effect of expression of Pfdn5 or Pfdn6 on memory performance. We found that pan-neuronal expression of either Pfdn5 (*Elav-Gal4/+; UAS-Pfdn5/+*: naïve, 18.27 ± 2.75 v/s trained, -4.43 ± 3.21; p < 0.001) or Pfdn6 (*Elav-Gal4/+; UAS-Pfdn6/+*: naïve, 18.37 ± 3.19 v/s trained, -0.88 ± 4.73; p < 0.001) does not cause any defect in naïve odor response or memory performance after training (Figure 7D-E [\[13\]](#)). However, coexpression of Pfdn5 significantly rescued the hTau^{V337M}-induced memory defects (*Elav-Gal4/UAS-hTau^{V337M}; UAS-*

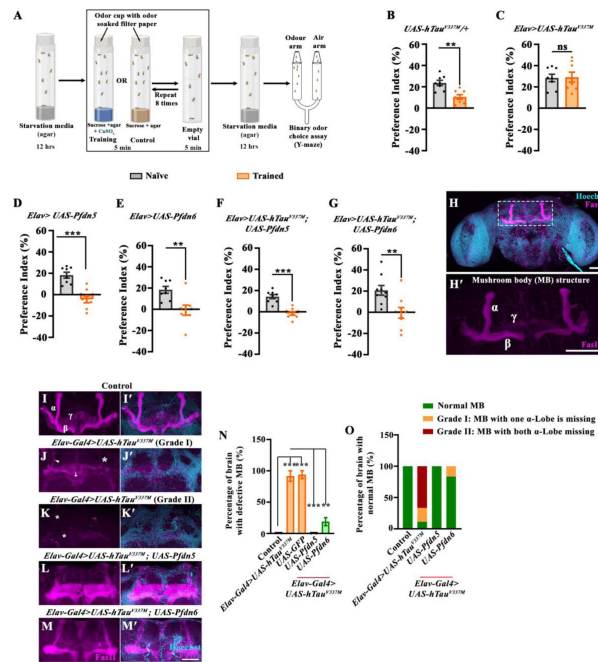


Figure 7

Over-expression of Pfdn5 or Pfdn6 rescues Tau-induced defects in learning and memory

(A) Cartoon of the Y-maze assay used for behavioural testing of conditioned odor preferences. Schematics of the protocol used to induce and measure a form of learning and memory (Mohandasan et al. 2022). During training, flies are exposed to a normally attractive odorant 2,3-butanedione (10^{-3} -fold dilution) in a spaced training protocol: 8X repeats of a training trial involving 5 minutes in the presence unpleasant medium (80 mM CuSO_4 + 85 mM sucrose and 0.75% agar) followed by 5 min in an air-filled empty vial. Trained flies were tested in a binary odor-choice assay in a Y-maze apparatus for their odor vs. air preference. Control flies were trained in 0.75 % agar media with 85 mM sucrose and similarly tested in Y-maze. **(B-G)** Histogram showing the quantification of odor preference index of naïve and trained flies towards 2,3-BD in Y-maze showing normal memory in control. A reduction in the preference index after training reflects levels of learning and memory **(B)** *UAS-hTau^{V337M}/+*; naïve flies (23.66 ± 2.42), trained flies (10.4 ± 2.14), whereas pan-neuronal expression of the pathological variant hTau causes a defect in long-term memory response **(C)** *Elav-Gal4/UAS-hTau^{V337M}*; naïve flies (28.39 ± 3.47), trained flies (29.13 ± 4.65). Notably, pan-neuronal overexpression of Pfdn5 **(D)** *Elav-Gal4/+; UAS-Pfdn5/+*; naïve flies (18.27 ± 2.75), trained flies (-4.43 ± 3.21), pan-neuronal overexpression of Pfdn6 **(E)** *Elav-Gal4/+; UAS-Pfdn6/+*; naïve flies (18.37 ± 3.19), trained flies (-0.88 ± 4.73) were normal. Interestingly, pan-neuronal over-expression of Pfdn5 along with hTau^{V337M} expression rescues the learning and memory deficits in **(F)** *Elav-Gal4/UAS-hTau^{V337M}; UAS-Pfdn5/+*; naïve flies (14.28 ± 1.96), trained flies (-1.5 ± 1.7). Consistently, pan-neuronal over-expression of Pfdn6 along with hTau^{V337M} expression also rescues the learning and memory deficits in **(G)** *Elav-Gal4/UAS-hTau^{V337M}; UAS-Pfdn6/+*; naïve flies (20.73 ± 4.58), trained flies (-0.52 ± 5.07). $n = 8$ biological replicates in each case. Error bars represent the standard error of the mean (SEM). $**p < 0.01$; $***p < 0.001$; ns, not significant. **(H)** Confocal image of *Drosophila* brain labeled with anti-FasII antibody (magenta) and Hoechst (cyan) showing mushroom body structure in wildtype animals. Scale bar 50 μm . **(H')** Confocal image of *Drosophila* mushroom body showing α -lobe, β -lobe and γ -lobe. **(I-M')** Confocal images showing mushroom body organisation in **(I-I')** control, **(J-J')** *Elav-Gal4>UAS-hTau^{V337M}* Grade I (Grade I represent the defective mushroom body with one α -lobe missing), **(K-K')** *Elav-Gal4>UAS-hTau^{V337M}* Grade II (Grade II represent the severe defects in mushroom body with both α -lobe missing), **(L-L')** *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5*, **(M-M')** *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6* double immunolabelled with FasII (magenta) and Hoechst (cyan). Scale bar in **(M')** for **(I-M')** represents 20 μm . The arrow points towards the β -lobe crossing midline, the arrowhead points to the thinner alpha lobe, and the asterisk represents the missing lobe. **(N)** Histogram showing the quantification for the percentage of *Drosophila* brain having defective mushroom body in control (0.00 ± 0.00), *Elav-Gal4>UAS-hTau^{V337M}* (91.67 ± 8.33), *Elav-Gal4>UAS-hTau^{V337M}; UAS-GFP* (93.75 ± 6.25), *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5* (0.00 ± 0.00), *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6* (18.75 ± 6.25). $***p < 0.001$; ns, not significant. The error bar represents the standard error of the mean (SEM); the statistical analysis was done using one-way ANOVA. **(O)** Histogram showing the quantification of mushroom body organization as normal mushroom bodies (NBs), Grade I or Grade II. In controls (NBs: 100, Grade I: 0, Grade II: 0), *Elav-Gal4>UAS-hTau^{V337M}* (NBs: 11.1, Grade I: 22.2%, Grade II: 66.7), *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn5* (NBs: 100, Grade I: 0, Grade II: 0), *Elav-Gal4>UAS-hTau^{V337M}; UAS-Pfdn6* (NBs: 83.3, Grade I: 16.6, Grade II: 0).

Pfdn5^{+/+}: naïve, 14.28 ± 1.96 v/s trained, -1.5 ± 1.7 ; $p < 0.001$) (**Figure 7F**). Similarly, co-expression of *Pfdn6* also significantly restored the *hTau*^{V337M}-induced memory defects (*Elav-Gal4/UAS-hTau*^{V337M}; *UAS-Pfdn6*^{+/+}: naïve, 20.73 ± 4.58 v/s trained, -0.52 ± 5.07 ; $p < 0.001$) (**Figure 7G**). Together, these data strengthen our observations that neuronal expression of *Pfdn5* or *Pfdn6* not only rescues Tau-induced neurodegeneration but also learning and memory deficits.

In *Drosophila*, the mushroom body is crucial for associative learning and memory. Mushroom body neuroblasts (MBNBs) produce Kenyon cells, which differentiate into three subtypes: α/β neurons, α'/β' neurons and γ neurons (Kunz et al. 2012) (**Figure 7H-H'**). Expression of *hTau* in *Drosophila* has been shown to disrupt MB architecture, predominantly affecting the α -lobe (Mershin et al. 2004). Given that overexpression of *Pfdn5* or *Pfdn6* rescues the learning and memory impairments associated with *hTau*^{V337M} expression, we examined whether *Pfdn5* or *Pfdn6* could prevent MB structural disruption. Immunostaining with FasII revealed severe MB abnormalities upon *hTau*^{V337M} expression (**Figure 7I-K**). Strikingly, coexpression of *Pfdn5* or *Pfdn6* completely restored MB integrity, effectively suppressing Tau-induced toxicity (**Figure 7I-O**). These findings indicate that *Pfdn5* and *Pfdn6* protect against Tauopathy-associated memory loss by maintaining the structural integrity of the mushroom body.

Cotranslational functions of *Pfdn5* do not completely explain its effects on neuronal microtubule stability, synapse morphology, and Tau-aggregation

Consistent with cell culture and biochemical studies (Tahmaz et al. 2021; Gestaut et al. 2022), *Drosophila* *Pfdn5* regulates tubulin monomers essential for microtubule assembly. Thus, one mechanism by which *Pfdn5* influences Tau, could be via its effect on tubulin levels. However, since *Pfdn5* colocalizes and binds neuronal microtubules, it could, alternatively, directly stabilize microtubules in axons and, by allowing Tau association with microtubules, prevent aggregation of free cytoplasmic Tau protein. To examine these models, we increased tubulin monomer levels in *Pfdn5* mutants by neuronally expressing α -*Tub* transgene and asked if it restored tubulin levels in the fly, and whether such restoration would be sufficient to rescue the neuronal microtubules and synaptic defects observed in *Pfdn5* mutants. Neuronal expression of α -*Tub* in *Pfdn5* mutant background restored both α - and β -tubulin monomers as well as ace-Tubulin to near wild-type levels (ace-Tubulin level: $\Delta Pfdn5^{15/40}$ (0.34 ± 0.13) vs *Elav-Gal4>UAS- α -Tubulin*; $\Delta Pfdn5^{15/40}$ (1.38 ± 0.15); $p < 0.01$) (**Figure 8A** and Figure S13A-D). However, this was insufficient to rescue the axonal microtubule level and organization (Tubulin intensity at synapses: $\Delta Pfdn5^{15/40}$ (0.18 ± 0.01), *Elav-Gal4>UAS- α -Tubulin*; $\Delta Pfdn5^{15/40}$ (0.23 ± 0.01); $p > 0.24$) (**Figure 8B-E**) or synaptic phenotypes associated with *Pfdn5* mutations (satellite boutons: $\Delta Pfdn5^{15/40}$ (15.5 ± 1.48), *Elav-Gal4>UAS- α -Tubulin*; $\Delta Pfdn5^{15/40}$ (14.25 ± 1.39); $p > 0.65$) (Figure S13E-I). Moreover, axonal *hTau* aggregates seen in neuronally expressing *hTau*^{V337M} in *Pfdn5* mutants were not reduced when tubulin monomer levels were restored (*hTau* punctae: *Elav-Gal4/UAS-hTau*^{V337M}; $\Delta Pfdn5^{15/40}$: 3.32 ± 0.65 / $100 \mu\text{m}^2$; vs *Elav-Gal4/UAS-hTau*^{V337M}; *UAS- α -tubulin*, $\Delta Pfdn5^{15/40}$: 2.9 ± 0.57 / $100 \mu\text{m}^2$; $p > 0.80$) (**Figure 8F-I**). These results prompted us to examine whether *Pfdn5* directly interacts with *hTau*. Although we observed colocalization of *Pfdn5* and *hTau* in axons (Figure S13J-K), co-immunoprecipitation did not detect a physical interaction between the two proteins under the conditions we performed the pull-down experiments. Thus, these data suggest that in addition to its role as a cochaperone for tubulin monomers, *Pfdn5* has an additional and potentially local role in stabilizing the neuronal microtubules as well as in preventing *hTau* aggregation.

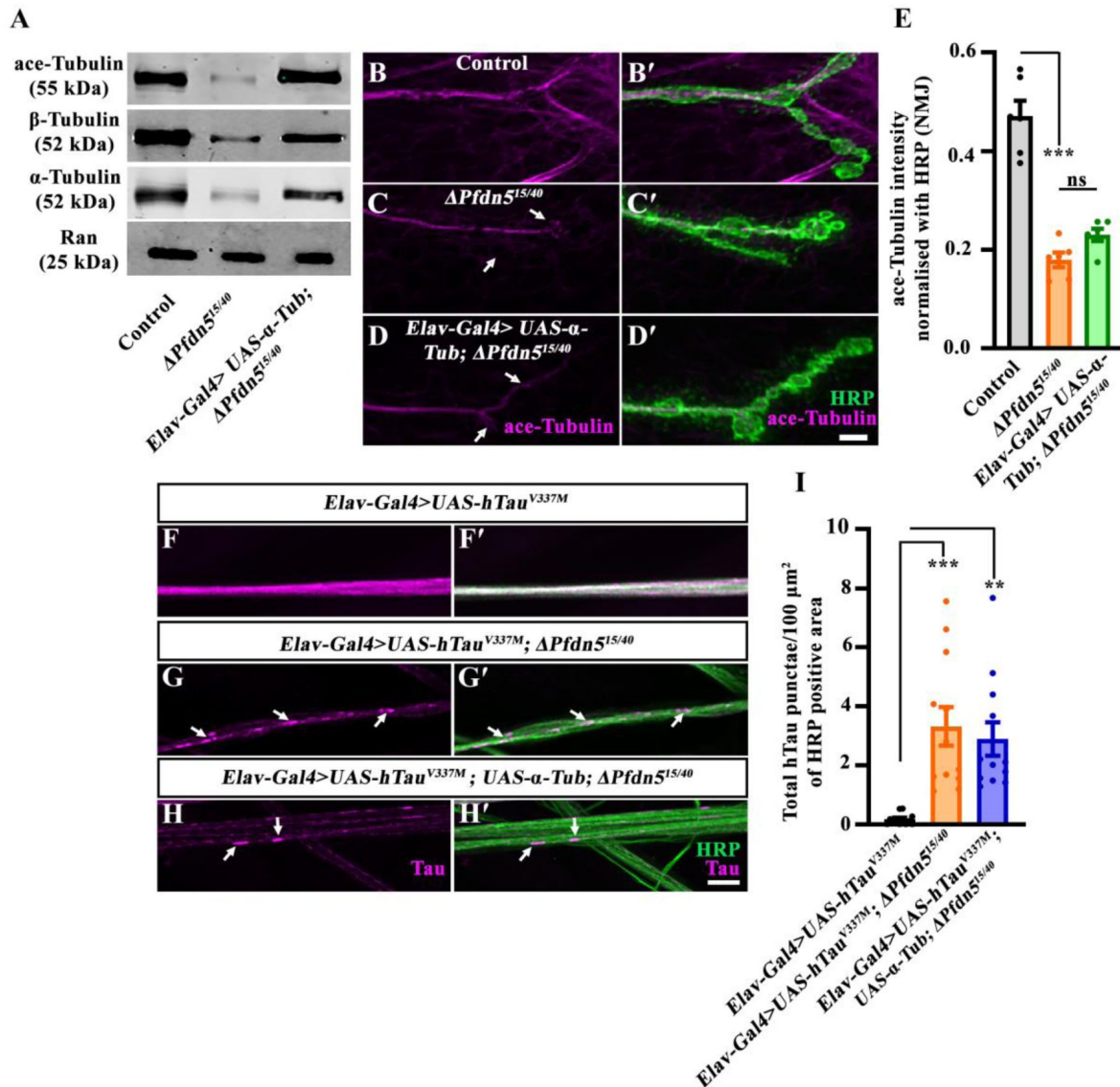


Figure 8

Microtubule stability and Tau-association with microtubules require Pfdn5 functions downstream of tubulin monomer expression

(A) Western blot showing protein levels of ace-Tubulin, α-Tubulin, and β-Tubulin in control, ΔPfdn5^{15/40}, and Elav-Gal4>UAS-α-Tubulin; ΔPfdn5^{15/40}. Ran protein levels were used as an internal loading control. **(B-D')** Confocal images of NMJ synapses showing synaptic microtubules in (B-B') control, (C-C') ΔPfdn5^{15/40}, (D-D') Elav-Gal4>UAS-α-Tubulin; ΔPfdn5^{15/40} double immunolabeled for ace-Tubulin (magenta) and HRP (green). The scale bar in D' for (A-D') represents 10 μm. Arrows in C and D shows disrupted microtubules. **(E)** Histogram showing ace-Tubulin intensity at the NMJ in control (0.47 ± 0.03), ΔPfdn5^{15/40} (0.18 ± 0.01), Elav-Gal4>UAS-α-Tubulin; ΔPfdn5^{15/40} (0.23 ± 0.01). ***p<0.001; ns, not significant. At least 6 NMJs of each genotype were used for quantification. **(F-H')** Confocal images of third instar larval axons in (F-F') Elav-Gal4>UAS-hTau^{V337M}, (G-G') Elav-Gal4>UAS-hTau^{V337M}; ΔPfdn5^{15/40}, and (H-H') Elav-Gal4>UAS-hTau^{V337M}; UAS-α-Tubulin, ΔPfdn5^{15/40} double immunolabeled with neuronal membrane marker, HRP (green), and T46 antibody against total human Tau (magenta). The scale bar in H' for (F-H') represents 10 μm. Arrows point to the Tau-aggregates. **(I)** Histogram showing the quantification of the number of Tau punctae per 100 μm² normalized with HRP positive area in Elav-Gal4>UAS-hTau^{V337M} (0.18 ± 0.05), Elav-Gal4>UAS-hTau^{V337M}; ΔPfdn5^{15/40} (3.32 ± 0.65), and Elav-Gal4>UAS-hTau^{V337M}; UAS-α-Tubulin, ΔPfdn5^{15/40} (2.9 ± 0.57). ***p<0.001; **p<0.01. At least 12 axons from 3 animals of each genotype were used for quantification.

Discussion

Through varied and detailed analyses performed in established *Drosophila* Tauopathy models, we identify Prefoldin as a crucial component of chaperone systems that mitigate hTau-aggregation-induced neurodegeneration. The experiments that lead to this conclusion provide three significant insights. First, that Prefoldin acts *in vivo* to suppress multiple measures of Tau-mediated degeneration. Second, the mechanism of Prefoldin action in Tau-toxicity goes beyond its established role in co-translational folding of monomeric tubulin. Third, and finally, that overexpression of Prefoldin is sufficient to delay the progression of Tau-toxicity *in vivo*. We consider each of these issues in turn below.

Prefoldin acts *in vivo* to suppress multiple measures of Tau-mediated degeneration

Seminal work by others in the field has both established the value of modeling Tauopathies in *Drosophila* and described a series of independent Tau-induced degenerative phenotypes displayed by these models (Shulman et al. 2014 [↗](#); Zhou et al. 2017 [↗](#); Vourkou et al. 2023 [↗](#); Bukhari et al. 2024 [↗](#)). The initial discovery that led to the rest of our current study was the identification of subunits of the prefoldin complex in a genetic screen for modifiers of Tau-toxicity. Knockdown of Pfdn components significantly enhanced eye-ommatidial degeneration in hTau^{V337M} expressing animals, suggestive of a role for this chaperone network in controlling the onset and progression of Tauopathies. Given the peripheral location of photoreceptors in the eye, it was important to more deeply assess the role of the identified chaperone components in the central nervous system. Such additional experiments confirmed that loss of Pfdn5 enhanced several additional hTau^{V337M} induced phenotypes, including synaptic organization. In addition, loss of Pfdn5 resulted in a striking increase in large Tau protein aggregates in larval axons as well as in the larval brain. These data demonstrate an essential role for Pfdn in restricting hTau toxicity *in vivo*. However, more dramatic was the observation that neuronal overexpression of either Pfdn5 or Pfdn6 was sufficient to mitigate hTau-induced brain vacuolization and memory decline. Together, these observations demonstrate a pivotal role for Pfdn, or at least its Pfdn5 and Pfdn6 subunits, in suppressing Tau-pathologies.

Mechanism of Prefoldin action in Tau-toxicity

Molecular analysis of FTDP-17/FTLD-tau mutations, as well as biochemical analysis of the pathogenic proteins, has shown that most disease-causing Tau mutations liberate Tau from microtubules and free the protein to form cytoplasmic aggregates (Hong et al. 1998 [↗](#); Dayanandan et al. 1999 [↗](#); Guo et al. 2017 [↗](#)). Therefore, a simple potential mechanism by which Pfdn5 influences Tau toxicity could be through its effect on reducing levels of tubulin monomer, which would be predicted to reduce the availability of stable microtubules and thereby liberate excess Tau to form potentially pathogenic aggregates in the cytosol. Consistent with this, studies in mice and *C. elegans* have shown that neurons with reduced tubulin levels are highly susceptible to the early onset of Tau-induced neuronal dysfunction (Tatebayashi et al. 2002 [↗](#); Yoshiyama et al. 2007 [↗](#); Miyasaka et al. 2018 [↗](#)). We noted that our screen for modulators of Tau-induced neurotoxicity also identified TBCE and the components of the CCT complex, which represent additional players of a chaperone network known to participate in the cotranslational folding of nascent actin or tubulin monomers (Hansen et al. 1999 [↗](#); Gil-Krzewska et al. 2010 [↗](#); Serna et al. 2015 [↗](#)). Further, our experiments showed that Pfdn5 mutations disrupt axonal microtubule organization as revealed by a reduction in the levels and organization of the microtubule-associated protein Futsch in axonal terminals of *Pfdn5* mutants (Figure 2 [↗](#)) (Roos et al. 2000a [↗](#); Sherwood et al. 2004 [↗](#); Jin et al. 2009 [↗](#)). And finally, a prior observation that the Tau-induced eye

degeneration is enhanced by the knockdown of TBCE has been proposed to be caused by perturbed microtubule dynamics in both *Drosophila* models and human patients (Dou et al. 2003 [↗](#); Fujiwara et al. 2020 [↗](#); Battini et al. 2021 [↗](#)).

Despite the above observations, the effect on tubulin monomer levels does not completely explain how Pfdn influences microtubule organization or Tau toxicity. First, we report the unexpected but robust observation that Pfdn5 is a microtubule-associated protein, physically associated with stable axonal microtubules, and, therefore, well positioned to directly influence microtubule stability (Figure 3 [↗](#)). Second and more directly, we find that genetic restoration of α -tubulin and β -tubulin monomers, as well as acetylated tubulin levels, was not sufficient to rescue the synaptic defects observed in *Pfdn5* mutants (Figure 8 [↗](#)).

Pfdn5 appears to influence Tau toxicity through a mechanism downstream of its known roles in cotranslational folding of tubulin. While this could be via its function as a novel microtubule-associated protein, an additional possibility we consider is that Pfdn, and by extension other known cotranslational chaperones, could act additionally and directly as holdases or disaggregases for Tau and/or other aggregation-prone proteins. There is considerable circumstantial evidence to indicate posttranslational and direct roles for Prefoldin, as well as CCT, in preventing the aggregation of misfolded proteins. For instance, Prefoldin not only inhibits the formation of larger Htt aggregates (Tashiro et al. 2013 [↗](#)) or amyloid β -aggregates (Sorgjerd et al. 2013 [↗](#)) but also solubilizes the amyloid oligomers and inhibits their fibril formation under *in vitro* conditions (Sakono et al. 2008 [↗](#); Sorgjerd et al. 2013 [↗](#)). Similarly, CCT/ TRiC complex physically associates with the polyQ repeats of Htt protein and remodels pathogenic aggregates *in vitro* (Tam et al. 2006 [↗](#); Darrow et al. 2015 [↗](#)). These evidence supports a direct role of Prefoldin and CCT as ‘disaggregase’ or ‘aggregate remodellar’ for aggregate-prone proteins and might regulate assembly/disassembly of Tau protein.

The identification of this cytoskeleton-regulatory chaperone network as a major modulator of Tauopathy supports the hypothesis that an age-dependent compromise in the chaperone activity could vitiate the onset and progression of multiple forms of Tauopathies and potentially other neurodegenerative diseases (Tittelmeier et al. 2020 [↗](#); Cyske et al. 2023 [↗](#)). Pfdn5 levels have been reported to decrease with age in a mouse model of Tauopathy (Kadoyama et al. 2019 [↗](#)); our findings provide direct evidence that even a minimal amount of pathological hTau in the absence of Pfdn5 could induce the early onset of hTau-induced neurodegeneration. Moreover, our finding that neuronal expression of α -tubulin rescues the hTau-induced synaptic defects in a manner that critically requires Pfdn5 activity extends the functional requirement of Prefoldins in the suppression of Tauopathies beyond their reported activity in cell culture or *in vitro* models (Millan-Zambrano et al. 2013 [↗](#); Sorgjerd et al. 2013 [↗](#); Tashiro et al. 2013 [↗](#); Takano et al. 2014 [↗](#)).

Prefoldin overexpression as a strategy to mitigate Tauopathies

Our work suggests that stabilizing the components of this chaperone system, particularly Pfdn5 or Pfdn6, could be a promising therapeutic approach for delaying Tau-induced neuropathology. Neuronal overexpression of Pfdn5 or Pfdn6 did not result in any detectable changes in synaptic morphogenesis or age-dependent neuronal degeneration. However, coexpression of Pfdn5 or Pfdn6 with the pathological variant of hTau remarkably suppressed Tau-induced synaptic defects, prevented brain vacuolization, and rescued memory defects in multiple forms of tauopathies (Figures 6 [↗](#) and 7 [↗](#)). This provides clear evidence that Pfdn5/Pfdn6-dependent microtubule regulation could potentially suppress Tau-induced neurodegeneration. These conclusions are supported by prior observations that expressing an acetylation mimic form of tubulin (Mao et al. 2017 [↗](#)) or stabilizing microtubules (Xiong et al. 2013 [↗](#)) rescues the synaptic defects induced in the *Drosophila* Tauopathy model.

Do Prefoldins have a general neuroprotective role? In neuronal cell lines, human Prefoldins colocalize with PolyQ-expanded protein Huntingtin and prevent the formation of toxic aggregates, supporting its role in the suppression of aggregation-induced neurotoxicity (Tashiro et al. 2013 [\[1\]](#)). A recent study has shown that age-dependent microtubule defects in *Drosophila* lead to dTau aggregation similar to aged human individuals, suggesting a crucial role of microtubule stability in tauopathies (Okenve-Ramos et al. 2024 [\[2\]](#)). Our findings that loss of Pfdn5 disrupts microtubules at an early stage and leads to Tau aggregation further support the crucial requirement of Pfdn5-dependent microtubule stability in suppressing various forms of Tauopathy. In addition, several lines of supportive evidence from human Alzheimer's disease datasets implicate PFDN5 in disease pathology. For example, recent compilations and analyses of proteomic data identified CCT components, TBCE, as well as Prefoldin subunits, including PFDN5, in AD tissue (Hsieh et al. 2019 [\[3\]](#); Tao et al. 2020 [\[4\]](#); Ji et al. 2022 [\[5\]](#); Askenazi et al. 2023 [\[6\]](#); Leitner et al. 2024 [\[7\]](#); Sun et al. 2024 [\[8\]](#)). Furthermore, whole blood mRNA expression data from Alzheimer's patients revealed downregulation of PFDN5 transcript (Ji et al. 2022 [\[5\]](#)). Together, these findings from human data support an essential role of PFDN5 in suppressing diverse neurodegenerative processes. Our data mechanistically extends these studies by revealing that Pfdn5 directly stabilizes neuronal microtubules, assists in proper partitioning of Tau onto the stable microtubules, and suppresses the formation of pathological Tau aggregates. Importantly, our data reveal that expression of Pfdn5, whether ubiquitous or neuron-specific, does not induce any observable synaptic or microtubule-associated defects in neurons. This finding holds significant therapeutic promise since modulating Pfdn5 or Pfdn6 expression or stability could safely and effectively mitigate neurodegenerative diseases associated with microtubule instability, such as Tauopathies and possibly FUS-induced neurodegeneration (Kandhavivorn et al. 2023 [\[9\]](#)).

Concluding remarks

Based on our findings, we propose a model in which Pfdn5 regulates microtubule formation and stability by two non-exclusive mechanisms: a) by regulating the folding of nascent tubulin monomers and b) by directly associating and stabilizing microtubules in neurons (**Figure 9** [\[10\]](#)). Both the functions of Pfdn5 are essential for regulating synaptic morphogenesis and Tau partitioning onto the microtubules. Normalizing the tubulin monomers to near wild-type levels was insufficient to rescue the axonal microtubule organization or suppress the Tau-aggregation in the absence of Pfdn5. This further supports the model that Pfdn5-dependent tubulin stabilization is essential for Tau partitioning and that Tau aggregation is microtubule-dependent. Thus, while the conventional chaperone function of Prefoldins is essential for tubulin folding, direct association of Pfdn5 with stable microtubules to limit its turnover in neurons is crucial for suppressing Tau-aggregation. Further elucidation of the underlying mechanisms of Pfdn5-mediated neuroprotection and chemical screens to identify novel small molecules may pave the way for novel strategies to preserve neuronal function and combat neurodegeneration.

Material and methods

Stocks and *Drosophila* husbandry

The *Drosophila* stocks were maintained at 25 °C in standard cornmeal medium containing sucrose, agar, and yeast granules. The larvae for experiments were grown at 25 °C in protein-rich media (80g/L cornflour, 40g/L dextrose, 20g/L sucrose, 18g/L agar, 15g/L yeast extract, 4% (v/v) propionic acid, 0.06% (v/v) ortho-phosphoric acid, and 0.07% methyl-4-hydroxy benzoate/Tego) under non-crowded conditions. The *w*¹¹¹⁸ was used as a control unless otherwise stated. All the genetic combinations and recombination were made using standard *Drosophila* genetics. The crosses for RNAi-mediated knockdown and the rescue experiments were grown at 25 °C. The following *Drosophila* lines were used in this study: *UAS-hTau*^{V337M} (Wittmann et al. 2001 [\[11\]](#)), *UAS-hTau*^{R406W}

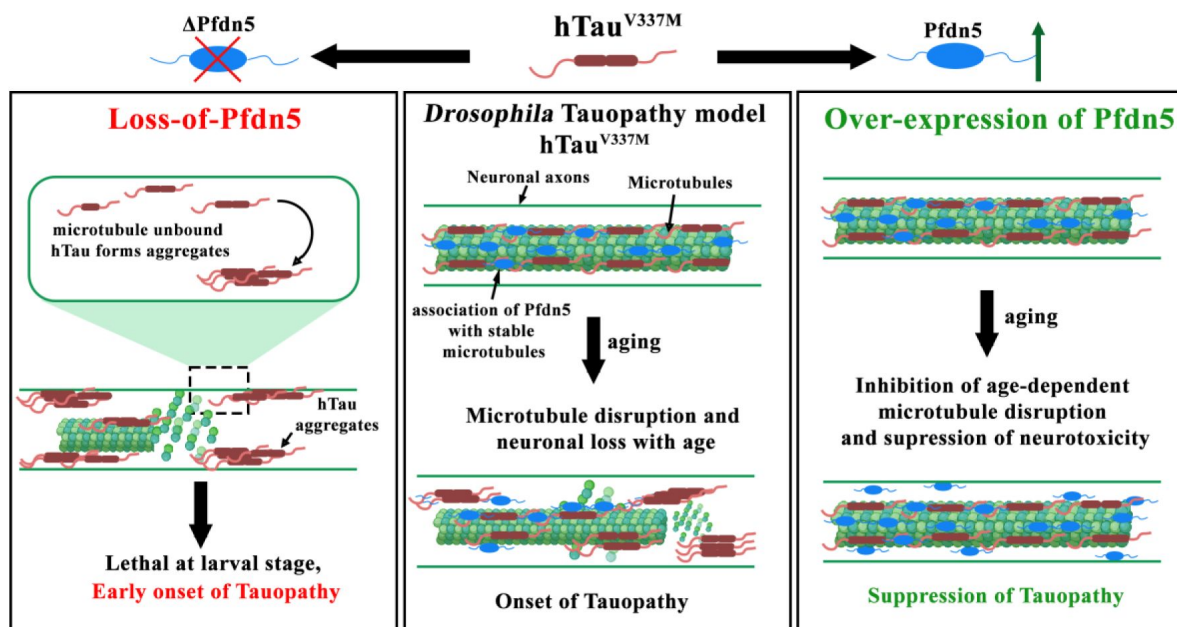


Figure 9

A model depicting novel functional requirement of Pfdn5 in microtubule stabilization and its role in suppressing age-dependent neuropathy.

In axons, Pfdn5 physically associates with microtubules and stabilizes them, thereby suppressing the turnover of microtubules. The pathological Tau dislodges from microtubules in an age-dependent manner and forms pathological aggregates that induce neuronal death (Middle panel). Loss of Pfdn5 disrupts neuronal microtubules, resulting in abnormal synaptic morphogenesis and facilitating the dislodging of microtubule-associated Tau, resulting in the formation of Tau-aggregates and stepping up the early onset of Tauopathies. An age-dependent reduction in the Pfdn5 levels or mutations in Pfdn5 could result in microtubule fragmentation and may facilitate Tau-induced neurotoxicity (Left panel). Pfdn5 suppresses Tau-aggregation in a manner that involves microtubule stability and does not appear to regulate Tau solubility directly. Notably, neuronal overexpression of Pfdn5 suppresses the microtubule disruption even in aged flies, thereby inhibiting the progression of Tauopathy (Right panel).

stocks were maintained at 25 °C in standard cornmeal medium containing sucrose, agar, and yeast granules. The larvae for experiments were grown at 25 °C in protein-rich media (80g/L cornflour, 40g/L dextrose, 20g/L sucrose, 18g/L agar, 15g/L yeast extract, 4% (v/v) propionic acid, 0.06% (v/v) ortho-phosphoric acid, and 0.07% methyl-4-hydroxy benzoate/Tego) under non-crowded conditions. The *w¹¹¹⁸* was used as a control unless otherwise stated. All the genetic combinations and recombination were made using standard *Drosophila* genetics. The crosses for RNAi-mediated knockdown and the rescue experiments were grown at 25 °C. The following *Drosophila* lines were used in this study: *UAS-hTau^{V337M}* (Wittmann et al. 2001 [\[1\]](#)), *UAS-hTau^{R406W}* (Wittmann et al. 2001 [\[1\]](#)), *UAS- α -Tub84B* (BL-7373), *actin5C-Gal4* (BL-25374); *Elav^{C155}-Gal4* (BL-458), *mef2-Gal4* (BL-50742), and *GMR-Gal4* (BL-9146). Details of RNAi lines used in this study are mentioned in Supplemental Table 1.

Scanning electron microscopy

The flies were immersed in fixative (1% glutaraldehyde, 1% formaldehyde, and 1M sodium cacodylate, pH 7.2) for 2 hr, followed by subsequent washes and dehydration via an ethanol series. The samples were then dried and sputter-coated as previously described (Choudhury et al. 2016 [\[2\]](#)). The flies were mounted on carbon conductive tabs stuck on aluminium stubs and imaged using a Zeiss scanning electron microscope (Carl Zeiss, Germany).

Generation of *Pfdn5* loss-of-function mutants and *Pfdn* transgenes

To generate the loss-of-function mutants of *Pfdn5*, two sets of gRNAs were designed for the *Pfdn5* genomic region using the CRISPR Optimal Target Finder online tool. The two gRNA pairs (gRNA1FP, gRNA1RP, and gRNA2FP, gRNA2RP) were cloned into a dual gRNA pCFD4 vector having a BbsI restriction site using Gibson Assembly Kit (New England Biolabs Ltd, UK) following the manufacturer's guidelines. The pCFD4 vector containing *Pfdn5* gRNAs was injected into *Drosophila* embryos to generate the transgene. Next, the transgenic flies containing the *Pfdn5* gRNAs were crossed with *nanos-Cas9* (BL-54591) to create the deletion of the *Pfdn5* gene in the germline cells. Following standard genetic crosses, lines were established in the F2 generation, and *Pfdn5* deletion was screened by PCR using primers *Pfdn5_FP1* and *Pfdn5_RP*. Two null mutants of *Pfdn5*, $\Delta Pfdn5^{15}$ (606 bp deletion), and $\Delta Pfdn5^{40}$ (577 bp deletion) were obtained and verified by sequencing using primer: *Pfdn5_Seq FP*.

To generate *Pfdn5* or *Pfdn6* transgenes, a full-length *Pfdn5* or *Pfdn6* ORF was amplified from cDNA and cloned into the Gal4-based expression vector pUAS at EcoRI and NotI restriction sites. The pUAS vector containing the *Pfdn5* or *Pfdn6* ORF was injected into *Drosophila* embryos to generate the transgene. Semiquantitative RT-PCR was used to assess the expression of the *Pfdn5* transcript in *Pfdn5* mutants. In brief, total RNA was isolated from larval fillets using TRIzol reagent (Invitrogen, Waltham, MA, USA). Reverse transcription was performed on 1 μ g total RNA using SuperscriptTM II Reverse Transcriptase (Invitrogen, Waltham, MA, USA) using an oligo-dT primer to make cDNA. The resulting cDNA was used for PCR to analyze the level of *Pfdn5* transcript using primers *Pfdn5_RTTP* and *Pfdn5_RTRP*. The list of primers used in this study is reported in Supplemental Table 2.

Generation of *Pfdn5* antibody

To generate antibodies against *Pfdn5*, the full-length *Pfdn5* was amplified from cDNA using primers *Pfdn5_pET28 FP* and *Pfdn5_pET RP* and cloned into the pET-28a (+) bacterial expression vector at NotI and EcoRI restriction sites. The His-tagged fusion protein was expressed in BL21 codon+ cells, purified from inclusion bodies using the standard protein purification method from the pellet fraction, and injected into mice (animal facility, IISER Bhopal). The antibody was used at a 1:200 dilution on fillets and a 1:5000 dilution for western blotting.

Quantitative RT-PCR

Total RNA was isolated from larval fillets (actin-Gal4>UAS-RNAi) using Qiagen RNA extraction kit, following the manufacturer's instructions (Invitrogen, Waltham, MA, USA). First-strand cDNA was synthesized using PrimeScript 1st strand cDNA Synthesis Kit (Takara, 6110A). Quantitative RT-PCR (qRT-PCR) reactions were set up using iTaq™ Universal SYBR® Green Supermix (#1725124, Bio-Rad) in the qTOWER³ (Analytik Jena, Jena, Germany) qPCR machine according to the manufacturer's protocol. The primers were designed using the IDT Primer Quest tool (<https://www.idtdna.com>) and are listed in Supplementary Table 2. *rp49* was used as an internal control. Three independent RT-qPCR runs were performed. The fold change was calculated using $2^{-\Delta(\Delta Ct)}$ (Livak and Schmittgen 2001).

Immunocytochemistry

Wandering third instar larvae were dissected on a sylgard plate in cold calcium-free HL3 and fixed in 4% paraformaldehyde in PBS for 30 minutes or in methanol for 5 minutes. The larval fillets were washed three times in PBS containing 0.2% Triton X-100, followed by blocking for one hour in 0.2% PBST containing 5% BSA. Fillets were fixed in the methanol for 5 mins to stain the acetyubulin in the muscles. The fillets were incubated overnight at 4°C with a primary antibody, followed by fluorophore-conjugated secondary antibodies at room temperature for 90 minutes. Finally, larval fillets were mounted on a glass slide with Fluoromount-G™ aqueous mounting medium (ThermoFisher, Waltham, MA, USA). Primary antibodies used in the study, mouse anti-CSP (ab49, 1:50), mouse anti-Futsch (22C10, 1:50), and mouse anti-β-tubulin (E7, 1:50) were obtained from the Developmental Studies Hybridoma Bank (University of Iowa, USA). Other primary antibodies used in this study are mouse anti-dPdn5 (this study, 1:200), mouse anti-acetyubulin (1:500, Sigma-Aldrich, Missouri, USA), anti-Tau (T46, 1:100, Invitrogen, Waltham, MA, USA), anti-FasII (1:50, DSHB), anti-D5D8N (1:500, CST, Boston, MA), and anti-phospho-Tau (AT8, 1:100, Invitrogen, Waltham, MA, USA). The fluorophore-conjugated secondary antibody Alexa Fluor 488 or Alexa Fluor 568 (Thermo Fisher Scientific, Waltham, MA, USA) was used at 1:800 dilution. Alexa Fluor 488 or Rhodamine conjugated anti-HRP (Jackson ImmunoResearch, Baltimore, PA, USA) were used at 1:800 dilution. Hoechst (Thermo Fisher Scientific, Waltham, MA, USA) was used at a 1:5000 dilution for 5 mins.

The brain staining for assessing vacuolization was done as previously described (Behnke et al. 2021). Briefly, the adult flies of appropriate genotypes were anesthetized and beheaded. The head was fixed in 4% PFA in 1X PBS containing 0.5% Triton X-100 for 20 min. The brain was dissected and fixed for another 2 hours, washed with PBST, and incubated with Hoechst (1:5000) and Alexa fluor 568 conjugated Phalloidin (1:100, Thermo Fisher Scientific, Waltham, MA, USA) cocktail in PBST for 24 hours. The brains were washed five times with PBST, followed by a final wash in 1X PBS for 30 minutes to remove the residual detergents or air sac, and mounted with Fluoromount-G™ aqueous mounting medium (Thermo Fisher Scientific, Waltham, MA, USA) on a glass slide for visualization.

Western blot analysis

Third instar larval body wall muscle or adult *Drosophila* heads were homogenized in 1X SDS lysis buffer (50 mM Tris-Cl, pH 6.8; 25 mM KCl; 2 mM EDTA; 0.3 M sucrose; 2% SDS), boiled, and centrifuged at 3000g. The protein concentration was quantified using bicinchoninic acid (BCA) Protein assay (Simpson 2008). The homogenized sample was then combined with an equal volume of 2× Laemmli buffer (50 mM Tris-HCl, pH 6.8; 2% SDS; 2% β-Mercaptoethanol; 0.1% Bromophenol blue and 10% glycerol). Subsequently, 25 µg of protein was separated on a 12% SDS-PAGE gel and transferred to a Hybond-LFP PVDF membrane (GE Healthcare, Illinois, USA). The membrane was blocked in 5% skimmed milk in 1X Tris-buffered saline (TBS) with 0.2% Tween-20 (0.2% TBST) for 1 hour at room temperature and then incubated overnight with primary antibody.

After washing with 0.2% TBST, the membrane was incubated with HRP-conjugated secondary antibody for 1 hour at room temperature. The primary antibodies used were: mouse anti-Pfdn5 (this study, 1:5000), rabbit anti- α -tubulin (1:3000, CST, Mumbai, India), mouse anti- β -tubulin (E7, 1:300, DSHB, University of Iowa, USA), mouse anti-ace-tubulin (1:5000, Sigma-Aldrich, St. Louis, Missouri, USA), anti-Tau (T46, 1:1000, Invitrogen, Waltham, MA, USA), anti-phospho-Tau (AT8, 1:1000, Invitrogen, Waltham, MA, USA), anti-GAPDH (1:5000), and mouse anti-Ran (1:2000, BD Biosciences, New Jersey, USA). Signals were detected using the LI-COR Odyssey imaging system (LI-COR Biosciences, Lincoln, USA).

In vivo microtubule-binding assay

The microtubule binding assay was performed as described previously (Feuillette et al. 2010 [\[1\]](#); Ando et al. 2016 [\[2\]](#)). Fifty heads from wild-type adult flies were collected and homogenized in 100 μ l of Buffer-C+ (50 mM (HEPES); pH 7.1, 1.0 mM MgCl₂, 1.0 mM EGTA, protease inhibitor cocktail (Roche, Basel, Switzerland), and phosphatase inhibitor cocktail in the presence of 20 μ M Taxol or 40 μ M Nocodazole diluted in dimethylsulfoxide (DMSO). Homogenized heads were centrifuged at 1,000 $\times$ g for 10 minutes, and an aliquot of the supernatant was subjected to western blotting as the “input fraction”. The remaining supernatant was layered onto a two-volume cushion of Buffer-C+ with 50% sucrose. After centrifugation at 100,000 $\times$ g for 30 minutes, one-third of the supernatant containing soluble tubulin was collected from the top of the tube as the cytosol fraction, and the pellet containing microtubule polymers and proteins bound to microtubules was resuspended in 100 μ l of SDS-Tris-Glycine sample buffer. Protein concentration in each fraction was measured using the BCA Protein Assay Kit. Equal amounts of protein were loaded onto each lane of Tris-Glycine gels and analyzed by western blotting using anti-Pfdn5 or anti-ace-tubulin antibodies.

Tau solubility assay

Tau solubility assay was performed as described in (Vourkou et al. 2023 [\[3\]](#)). Briefly, adult fly heads were homogenized in TBS/sucrose buffer (50 mM Tris HCl, pH 7.4, 175 mM NaCl, 1M sucrose, 5 mM EDTA) supplemented with protease and phosphatase inhibitors. The homogenate was initially centrifuged at 1,000 $\times$ g for 2 min to remove debris. The resulting supernatant was subjected to ultracentrifugation at 200,000 $\times$ g for 2h at 4°C to separate soluble proteins. This supernatant comprised the soluble Tau fraction. The pellet, containing insoluble material, was resuspended in 5% SDS/TBS buffer and centrifuged again at 200,000 $\times$ g for 2h at 25°C. The resulting supernatant was collected as the SDS-soluble, aqueous-insoluble fraction, which was enriched for aggregated Tau. All fractions were diluted in Laemmli buffer, boiled, and resolved using SDS-PAGE. Immunoblotting with Tau-specific antibodies was used to analyze Tau species in the different fractions (Vourkou et al. 2023 [\[3\]](#)).

Drug treatments

Third instar *Drosophila* larvae were dissected in Schneider’s *Drosophila* Medium (Gibco, CA, USA). Following dissection, preparations were gently washed with fresh Schneider’s medium to remove residual debris. The preparations were incubated in Schneider’s medium containing 1,6-Hexanediol (1,6-HD; H6703, Sigma-Aldrich) at final concentrations of 0%, 1%, or 5% for 2 mins at 25 °C. The 0% condition consisted of Schneider’s medium alone (Liu et al. 2021 [\[4\]](#)). Following treatment, samples were immediately fixed in 4% paraformaldehyde in PBS for 30 minutes at room temperature and processed for immunostaining. For the LiCl treatments, all the genotypes were raised on the standard fly media containing zero mM or 20 mM LiCl (Cowan et al. 2010 [\[5\]](#)). The zero mM solution contained only DMSO.

Memory paradigm for aversive associative olfactory conditioning

To induce long-term aversive conditioning memory (LTM), flies were trained to associate an attractive odorant with bitter food, CuSO₄, as described previously (Mohandasan et al. 2022 [\[6\]](#)). 4–5-day-old adult flies were trained on 0.75% agar media containing 85 mM sucrose and 80 mM

CuSO₄ (punishment media); the same media without CuSO₄ was used as control media. Flies of specific genotypes were first starved in glass vials overnight containing 0.75% agar (starvation media) and then transferred to punishment media vials. For delivering the odor, a filter paper (1.5 cm x 2 cm), soaked in 100 µl of 5% 2,3 BD (2,3 butanedione, attractive odorant), was placed in a porous odor cup fitted at the top of the punishment or control vials. Starved flies were transferred into the punishment or control vials for 5 minutes, followed by 5 minutes of incubation in an empty test tube. This training cycle was repeated eight times for both the punishment and control groups of flies. For checking 1-day memory retention, flies are starved for 6 hours after the 8-cycle conditioning step, followed by a 5-minute food pulse and again starvation for 18 hours. The flies were then tested (24 hrs after training) for their preference towards 2,3 BD in a binary odor choice assay paradigm using a Y-maze. The Preference Index (PI) of the control and trained flies was calculated below.

$$\text{Preference Index} = \frac{\text{Odor Arm Flies} - \text{Air Arm Flies}}{\text{Odor Arm Flies} + \text{Air Arm Flies}} \times 100$$

Quantifications and statistical analysis

For bouton quantification, images were captured with a laser scanning confocal microscope (FV3000; Olympus) using 40x 1.3 NA or 60x 1.42 NA objectives and processed using ImageJ (National Institutes of Health, USA) or Adobe Photoshop software (Adobe Inc., USA). NMJs from muscle 4 at A2 hemisegment were captured using a 60x 1.42 NA objective to calculate the bouton number. CSP-positive boutons were counted manually. For bouton area quantification, NMJs from muscle 4 at A2 hemisegment were captured, and the area of five terminal boutons was calculated by drawing a free-hand sketch around CSP-positive boutons. The control and experimental fillets were processed similarly for fluorescence quantification, and the fluorescence images were captured under the same settings for every experimental set. For quantification of AT8 and T46 levels in the larval axons, HRP-marked boundaries were defined for each axon. The fluorescence intensity of AT8 or T46 was calculated and normalized with HRP fluorescence. To quantify the Tau punctae in the axons, z-projections of confocal images of third-instar larval axons were captured. T46 and AT8 positive punctae were manually counted and normalized with the area of respective axons. To quantify the Tau punctae in the larval brain, fluorescence threshold was set and analysed using ImageJ. Tau punctae greater than $> 3 \mu\text{m}^2$ were quantified.

For bright field imaging of eyes, flies were anesthetized using Diethyl ether (Sigma-Aldrich, Missouri, USA), and images were captured using Leica M205FA (Leica, Germany) Stereo Zoom Microscope. The percentage of the degenerated area was quantified as the area of the eyes showing roughness (for bright-field images) and the area containing fused ommatidia (for SEM images), normalized with the total area of the eye multiplied by 100. The percentage of fused ommatidia was quantified from the SEM images as the number of fused ommatidia normalized with the total ommatidia multiplied by 100. The maximum area of individual vacuoles was defined using the Wand tool in ImageJ software to quantify the vacuole size. Subsequently, the traced vacuoles were assigned and saved as regions of interest (ROIs). The selected ROIs were stacked and measured to quantify the size of the vacuoles (Behnke et al. 2021 [\[1\]](#)). The total number of boutons with Futsch-positive loops was quantified manually using ImageJ (Coyne et al. 2014 [\[2\]](#)).

Colocalization analysis was performed using the JACoP ImageJ plugin (Bolte and Cordelières 2006 [\[3\]](#)). A line was drawn across the axons, and plot profiles were drawn using the ImageJ function Plot Profile. The density of western blot bands was quantified using ImageJ software. For multiple comparisons, one-way ANOVA followed by post hoc Tukey's test or Student's t-test was used. GraphPad Prism 8 (GraphPad Software Inc., California, USA) was used to plot all the graphs. Error bars in all the histograms represent $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Data availability

All data associated with this study is included within the manuscript.

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Additional files

Supplemental Figures [↗](#)

Supplemental Table 1 [↗](#)

Supplemental Table 2 [↗](#)

Supplemental Table 3 [↗](#)

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Reviewer #1 (Public review):

Summary:

In this manuscript, Bisht et al address the hypothesis that protein folding chaperones may be implicated in aggregopathies and in particular Tau aggregation, as a means to identify novel therapeutic routes for these largely neurodegenerative conditions.

The authors conducted a genetic screen in the *Drosophila* eye, which facilitates identification of mutations that either enhance or suppress a visible disturbance in the nearly crystalline organization of the compound eye. They screened by RNA-interference all 64 known *Drosophila* chaperones and revealed that mutations in 20 of them exaggerate the Tau-dependent phenotype, while 15 ameliorated it. The enhancer of degeneration group included 2 subunits of the typically heterohexameric prefoldin complex and other co-translational chaperones.

The authors characterized in depth one of the prefoldin subunits, Pfdn5 and convincingly demonstrated that this protein functions in regulation of microtubule organization, likely due to its regulation of proper folding of tubulin monomers. They demonstrate convincingly using both immunohistochemistry in larval motor neurons and microtubule binding assays that Pfdn5 is a bona fide microtubule associated protein contributing to the stability of the axonal microtubule cytoskeleton, which is significantly disrupted in the mutants.

Similar phenotypes were observed in larvae expressing the Frontotemporal dementia with Parkinsonism on chromosome 17-associated mutations of the human Tau gene V377M and R406W. On the strength of the phenotypic evidence and the enhancement of the TauV377M-induced eye degeneration they demonstrate that loss of Pfdn5 exaggerates the synaptic deficits upon expression of the Tau mutants. Conversely, overexpression of Pfdn5 or Pfdn6 ameliorates the synaptic phenotypes in the larvae, the vacuolization phenotypes in the adult, even memory defects upon TauV377M expression.

Strengths:

The phenotypic analyses of the mutant and its interactions with TauV377M at the cell biological, histological, and behavioral levels are precise, extensive, and convincing and achieve the aims of characterization of a novel function of Pfdn5.

Regarding this memory defect upon V377M tau expression. Kosmidis et al (2010) PMID: 20071510, demonstrated that pan-neuronal expression of TauV377M disrupts the organization of the mushroom bodies, the seat of long-term memory in odor/shock and odor/reward conditioning. If the novel memory assay the authors use depends on the adult brain structures, then the memory deficit can be explained in this manner.

If the mushroom bodies are defective upon TauV377M expression does overexpression of Pfdn5 or 6 reverse this deficit? This would argue strongly in favor of the microtubule stabilization explanation.

The discovery that Pfdn5 (and 6 most likely) affect tauV377M toxicity is indeed a novel and important discovery for the Tauopathies field. It is important to determine whether this interaction affects only the FTDP-17-linked mutations, or also WT Tau isoforms, which are linked to the rest of the Tauopathies. Also, insights on the mode(s) that Pfdn5/6 affect Tau toxicity, such as some of the suggestions above are aiming at, will likely be helpful towards therapeutic interventions.

Weaknesses:

What is unclear however is how Pfdn5 loss or even overexpression affects the pathological Tau phenotypes.

Does Pfdn5 (or 6) interact directly with TauV377M? Colocalization within tissues is a start, but immunoprecipitations would provide additional independent evidence that this is so.

Does Pfdn5 loss exacerbate TauV377M phenotypes because it destabilizes microtubules, which are already at least partially destabilized by Tau expression?

Rescue of the phenotypes by overexpression of Pfdn5 agrees with this notion.

However, Cowan et al (2010) pmid: 20617325 demonstrated that wild-type Tau accumulation in larval motor neurons indeed destabilizes microtubules in a Tau phosphorylation-dependent manner.

So, is TauV377M hyperphosphorylated in the larvae?? What happens to TauV377M phosphorylation when Pfdn5 is missing and presumably more Tau is soluble and subject to hyperphosphorylation as predicted by the above?

Expression of WT human Tau (which is associated with most common Tauopathies other than FTDP-17) as Cowan et al suggest has significant effects on microtubule stability, but such Tau-expressing larvae are largely viable. Will one mutant copy of the Pfdn5 knockout enhance the phenotype of these larvae?? Will it result in lethality? Such data will serve to generalize the effects of Pfdn5 beyond the two FTDP-17 mutations utilized.

Does the loss of Pfdn5 affect TauV377M (and WTTau) levels?? Could the loss of Pfdn5 simply result in increased Tau levels? And conversely, does overexpression of Pfdn5 or 6 reduce Tau levels?? This would explain the enhancement and suppression of TauV377M (and possibly WT Tau) phenotypes. It is an easily addressed, trivial explanation at the observational level, which if true begs for a distinct mechanistic approach.

Finally, the authors argue that TauV377M forms aggregates in the larval brain based on large puncta observed especially upon loss of Pfdn5. This may be so, but protocols are available to validate this molecularly the presence of insoluble Tau aggregates (for example, pmid: 36868851) or soluble Tau oligomers as these apparently differentially affect Tau toxicity. Does Pfdn5 loss exaggerate the toxic oligomers and overexpression promotes the more benign large aggregates??

Comments on revisions:

In the revised manuscript Bisht et al have provided extensive new experimental evidence in support of previously more tenuous claims. These fully satisfy my comments and suggestions, and in my view, have significantly strengthened the manuscript with compelling new evidence.

<https://doi.org/10.7554/eLife.104691.2.sa2>

Reviewer #2 (Public review):

Bisht et al detail a novel interaction between the chaperone, Prefoldin 5, microtubules, and tau-mediated neurodegeneration, with potential relevance for Alzheimer's disease and other tauopathies. Using *Drosophila*, the study shows that Pfdn5 is a microtubule-associated protein, which regulates tubulin monomer levels and can stabilize microtubule filaments in the axons of peripheral nerves. The work further suggests that Pfdn5/6 may antagonize Tau aggregation and neurotoxicity. While the overall findings may be of interest to those investigating the axonal and synaptic cytoskeleton, the detailed mechanisms for the observed phenotypes remain unresolved and the translational relevance for tauopathy pathogenesis is yet to be established. Further, a number of key controls and important experiments are missing that are needed to fully interpret the findings.

The strength of this study is the data showing that Pfdn5 localizes to axonal microtubules and the loss-of-function phenotypic analysis revealing disrupted synaptic bouton morphology. The major weakness relates to the experiments and claims of interactions with Tau-mediated neurodegeneration. In particular, it is unclear whether knockdown of Pfdn5 may cause eye phenotypes independent of Tau. Further, the GMR>tau phenotype appears to have been incorrectly utilized to examine age-dependent, neurodegeneration.

This manuscript argues that its findings may be relevant to thinking about mechanisms and therapies applicable to tauopathies; however, this is premature given that many questions remain about the interactions from *Drosophila*, the detailed mechanisms remain unresolved, and absent evidence that tau and Pfdn may similarly interact in the mammalian neuronal context. Therefore, this work would be strongly enhanced by experiments in human or murine neuronal culture or supportive evidence from analyses of human data.

Comments on revisions:

The revision adequately addresses most of the previously raised concerns, resulting in a significantly improved manuscript.

<https://doi.org/10.7554/eLife.104691.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

In this manuscript, Bisht et al address the hypothesis that protein folding chaperones may be implicated in aggregopathies and in particular Tau aggregation, as a means to identify novel therapeutic routes for these largely neurodegenerative conditions.

*The authors conducted a genetic screen in the *Drosophila* eye, which facilitates the identification of mutations that either enhance or suppress a visible disturbance in the nearly crystalline organization of the compound eye. They screened by RNA interference all 64 known *Drosophila* chaperones and revealed that mutations in 20 of them exaggerate the Tau-dependent phenotype, while 15 ameliorated it. The enhancer of the degeneration group included 2 subunits of the typically heterohexameric prefoldin complex and other co-translational chaperones.*

The authors characterized in depth one of the prefoldin subunits, Pfdn5, and convincingly demonstrated that this protein functions in the regulation of microtubule organization, likely due to its regulation of proper folding of tubulin monomers. They demonstrate convincingly using both immunohistochemistry in larval motor neurons and microtubule binding assays that Pfdn5 is a bona fide microtubule-associated protein contributing to the stability of the axonal microtubule cytoskeleton, which is significantly disrupted in the mutants.

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(1) If the mushroom bodies are defective upon Tau^{V377M} expression, does overexpression of Pfdn5 or 6 reverse this deficit? This would argue strongly in favor of the microtubule stabilization explanation.

We thank the reviewer for this insightful comment. Consistent with Kosmidis et al. (2010), we confirm that expression of hTau^{V377M} disrupts the architecture of mushroom bodies. In addition, we find, as suggested by the reviewer, that coexpression of either Pfdn5 or Pfdn6 with hTau^{V377M} significantly restores the organization of the mushroom bodies. These new findings strongly support the hypothesis that Pfdn5 or Pfdn6 mitigate hTau^{V377M}-induced memory deficits by preserving the structure of the mushroom body, likely through stabilizing the microtubule network. This data has now been included in the revised manuscript (Figure 7H-O).

(2) The discovery that Pfdn5 (and 6 most likely) affects tauV377M toxicity is indeed a novel and important discovery for the Tauopathies field. It is important to determine whether this interaction affects only the FTDP-17-linked mutations or also WT Tau isoforms, which are linked to the rest of the Tauopathies. Also, insights on the mode(s) that Pfdn5/6 affect Tau toxicity, such as some of the suggestions above, are aiming at will likely be helpful towards therapeutic interventions.

We agree that determining whether prefoldin modulates the toxicity of both mutant and wildtype Tau is critical for understanding its broader relevance to Tauopathies. We have now performed additional experiments required to address this issue. These new data show that loss of Pfdn5 also exacerbates toxicity associated with wildtype Tau (hTau^{WT}), in a manner similar to that observed with hTau^{V377M} or hTau^{R406W}. Specifically, overexpression of hTau^{WT} in a Pfdn5 mutant background leads to Tau aggregate formation (Figure S7G-I), and coexpression of Pfdn5 with hTau^{WT} reduces the associated synaptic defects (Figure S11F-L). These findings underscore a general role for Pfdn5 in modulating diverse Tauopathy-associated phenotypes and suggest that it could be a broadly relevant therapeutic target.

Weakness

(3) What is unclear, however, is how Pfdn5 loss or even overexpression affects the pathological Tau phenotypes. Does Pfdn5 (or 6) interact directly with TauV377M? Colocalization within tissues is a start, but immunoprecipitations would provide additional independent evidence that this is so.

We appreciate this important suggestion. To investigate a potential direct interaction between Pfdn5 and Tau^{V377M}, we performed co-immunoprecipitation experiments using lysates from adult fly brain expressing hTau^{V377M}. Under the conditions tested, we did not detect a direct physical interaction. While this does not support a direct interaction, it does not strongly refute it either. We note that Pfdn5 and Tau are colocalized within axons (Figure S13J-K). At this stage, we are unable to resolve the issue of direct vs indirect association. If indirect, then Tau and Pfdn5 act within the same subcellular compartments (axon); if direct, then either only a small fraction of the total cellular proteins is in the Tau-Pfdn5 complex and therefore difficult to detect in bulk protein westerns, or the interactions are dynamic or occur in conditions that we have not been able to mimic in vitro.

(4) Does Pfdn5 loss exacerbate Tau^{V377M} phenotypes because it destabilizes microtubules, which are already at least partially destabilized by Tau expression? Rescue of the phenotypes by overexpression of Pfdn5 agrees with this notion.

However, Cowan et al (2010) PMID: 20617325 demonstrated that wildtype Tau accumulation in larval motor neurons indeed destabilizes microtubules in a Tau phosphorylation-dependent manner. So, is Tau^{V377M} hyperphosphorylated in the larvae?? What happens to Tau^{V377M} phosphorylation when Pfdn5 is missing and presumably more Tau is soluble and subject to hyperphosphorylation as predicted by the above?

We completely agree that it is important to link Tau-induced phenotypes with the microtubule destabilization and phosphorylation state of Tau. We performed immunostaining using futsch antibody to check the microtubule organization at the NMJ and observed a severe reduction in futsch intensity when Tau^{V377M} was expressed in the Pfdn5 mutant (ElavGal4>Tau^{V377M}; DPfdn5^{15/40}), suggesting that Pfdn5 absence exacerbates the hTau^{V377M} defects due to more microtubule destabilization (Figure S6F-J).

We have performed additional experiments to examine the phosphorylation state of hTau in Drosophila larval axons. Immunocytochemistry indicated that only a subset of hTau aggregates in Pfdn5 mutants (Elav-Gal4>Tau^{V377M}; DPfdn5^{15/40}) are recognized by phospho-hTau antibodies. For instance, the AT8 antibody (targeting pSer202/pThr205) (Goedert et al., 1995) labelled only a subset of aggregates identified by the total hTau antibody (D5D8N) (Figure S9AE). Moreover, feeding these larvae (Elav-Gal4>Tau^{V377M}; DPfdn5^{15/40}) with LiCl, which blocks GSK3b, still showed robust Tau aggregation (Figure S9F-J).

These results imply that: a) soluble phospho-hTau levels in Pfdn5 mutants are low and not reliably detected with a single phosphorylation-specific antibody; b) Loss of Pfdn5 results in Tau aggregation in a hyperphosphorylation-independent manner similar to what has been reported earlier (LI et al. 2022); and c) the destabilization of microtubules in Elav-Gal4>Tau^{V377M}; DPfdn5^{15/40} results in Tau dissociation and aggregate formation. These data and conclusions have been incorporated into the revised manuscript.

(5) Expression of WT human Tau (which is associated with most common Tauopathies other than FTDP-17) as Cowan et al suggest has significant effects on microtubule stability, but such Tauexpressing larvae are largely viable. Will one mutant copy of the Pfdn5 knockout enhance the phenotype of these larvae?? Will it result in lethality? Such data will serve to generalize the effects of Pfdn5 beyond the two FTDP-17 mutations utilized.

We have now examined whether heterozygous loss of Pfdn5 (Δ Pfdn5/+) enhances the effect of Tau expression. While each genotype (hTau^{V377M}, hTau^{WT} or Δ Pfdn5/+) alone is viable, Elav-Gal4 driven expression of hTau^{V377M} or hTau^{WT} in Pfdn5 heterozygous background does not cause lethality.

(6) Does the loss of Pfdn5 affect TauV377M (and WTTau) levels?? Could the loss of Pfdn5 simply result in increased Tau levels? And conversely, does overexpression of Pfdn5 or 6 reduce Tau levels?? This would explain the enhancement and suppression of Tau^{V377M} (and possibly WT Tau) phenotypes. It is an easily addressed, trivial explanation at the observational level, which, if true, begs for a distinct mechanistic approach.

To test whether Pfdn5 modulates Tau phenotypes by altering Tau protein levels, we performed western blot analysis under Pfdn5 or Pfdn6 overexpression conditions and observed no change in hTau^{V377M} levels (Figure 6O). However, in the absence of Pfdn5, both

hTau^{V337M} and hTau^{WT} form large, insoluble aggregates that are not detected in soluble lysates by standard western blotting but are visualized by immunocytochemistry (Figure S7G-I). Thus, the apparent reduction in Tau levels on western blots reflects a solubility shift, not an actual decrease in Tau expression. These findings argue against a simple model in which Pfdn5 regulates Tau abundance and instead support a mechanism in which Pfdn5 loss leads to change in Tau conformation, leading to its sequestration away from already destabilized microtubules.

(7) Finally, the authors argue that Tau^{V337M} forms aggregates in the larval brain based on large puncta observed especially upon loss of Pfdn5. This may be so, but protocols are available to validate this molecularly the presence of insoluble Tau aggregates (for example, pmid: 36868851) or soluble Tau oligomers, as these apparently differentially affect Tau toxicity. Does Pfdn5 loss exaggerate the toxic oligomers, and overexpression promote the more benign large aggregates??

We have performed additional experiments to analyze the nature of these aggregates using 1,6-HD. The 1,6-hexanediol can dissolve the Tau aggregate seeds formed by Tau droplets, but cannot dissolve the stable Tau aggregates (WEGMANN et al. 2018). We observed that 5% 1,6hexanediol failed to dissolve these Tau aggregates (Figure S8), demonstrating the formation of stable filamentous flame-shaped NFT-like aggregates in the absence of Pfdn5 (Figure 5D and Figure S9).

Reviewer #2 (Public review):

Bisht et al detail a novel interaction between the chaperone, Prefoldin 5, microtubules, and tau-mediated neurodegeneration, with potential relevance for Alzheimer's disease and other tauopathies. Using Drosophila, the study shows that Pfdn5 is a microtubule-associated protein, which regulates tubulin monomer levels and can stabilize microtubule filaments in the axons of peripheral nerves. The work further suggests that Pfdn5/6 may antagonize Tau aggregation and neurotoxicity. While the overall findings may be of interest to those investigating the axonal and synaptic cytoskeleton, the detailed mechanisms for the observed phenotypes remain unresolved and the translational relevance for tauopathy pathogenesis is yet to be established. Further, a number of key controls and important experiments are missing that are needed to fully interpret the findings.

The strength of this study is the data showing that Pfdn5 localizes to axonal microtubules and the loss-of-function phenotypic analysis revealing disrupted synaptic bouton morphology. The major weakness relates to the experiments and claims of interactions with Tau-mediated neurodegeneration.

In particular, it is unclear whether knockdown of Pfdn5 may cause eye phenotypes independent of Tau.

Our new experiments confirm that knockdown of Pfdn5 alone does not cause eye phenotypes.

Further, the GMR>tau phenotype appears to have been incorrectly utilized to examine agedependent, neurodegeneration.

In response, we have modulated and explained our conclusions in this regard as described later in our "rebuttal."

This manuscript argues that its findings may be relevant to thinking about mechanisms and therapies applicable to tauopathies; however, this is premature given that many

questions remain about the interactions from Drosophila, the detailed mechanisms remain unresolved, and absent evidence that Tau and Pfdn may similarly interact in the mammalian neuronal context. Therefore, this work would be strongly enhanced by experiments in human or murine neuronal culture or supportive evidence from analyses of human data.

The reviewer is correct that the impact would be greater if Pfdn5-Tau interactions were also examined in human tissue. While we have not attempted these experiments ourselves, we hope that our observations will stimulate others to test the conservation of phenomena we describe. There are, however, several lines of circumstantial evidence from human Alzheimer's disease datasets that implicate PFDN5 in disease pathology. For example, recent compilations and analyses of proteomic data show reductions of CCT components, TBCE, as well as Prefoldin subunits, including PFDN5, in AD tissue (HSIEH et al. 2019; TAO et al. 2020; JI et al. 2022; ASKENAZI et al. 2023; LEITNER et al. 2024; SUN et al. 2024). Furthermore, whole blood mRNA expression data from Alzheimer's patients revealed downregulation of PFDN5 transcript (JI et al. 2022). Together, these findings from human data are consistent with the roles of PFDN5 in suppressing diverse neurodegenerative processes. We have incorporated these points into the discussion section of the revised manuscript.

Reviewer #1 (Recommendations for the authors):

See public review for experimental recommendations focusing on the Tau Pfdn interactions. I would refrain from using the word aggregates, I would call them puncta, unless there is molecular or visual (ie AFM) evidence that they are indeed insoluble aggregates. Finally, although including the full genotypes written out below the axis in the bar graphs is appreciated, it nevertheless makes them difficult to read due to crowding in most cases and somewhat distracting from the figure.

In my opinion, a more reader-friendly manner of reporting the phenotypes will be highly helpful. For example, listing each component of the genotype on the left of each bar graph and adding a cross or a filled circle under the bar to inform of the full genotype of the animals used.

As described in the response to the previous comment, we now have strong direct evidences to support our view that the observed puncta are stable Tau aggregates. Thus, we feel justified to use the term Tau-aggregates in preference to Tau puncta.

We have tried to write the genotypes to make them more reader-friendly.

Reviewer #2 (Recommendations for the authors):

(1) Lines 119-121: 35 modifiers from 64 seem like an unusually high hit rate. Are these individual genes or lines? Were all modifiers supported by at least 2 independent RNAi strains targeting non-overlapping sequences? A supplemental table should be included detailing all genes and specific strains tested, with corresponding results.

We agree with the reviewer that 35 modifiers from 64 genes may be too high. However, since the genes knocked down in the study are chaperones, crucial for maintaining proteostasis, we may have got unusually high hits. The information related to individual genes and lines is provided in Supplemental Table 1. We have now included an additional Supplemental Table 3, which lists the genes and the RNAi lines used in Figure 1, detailing the sequence target information. The table also specifies the number of independent RNAi strains used and the corresponding results.

(2) Figure 1: The authors quantify the areas of ommatidial fusion and necrosis as degeneration, but it is difficult to appreciate the aberrations in the photos provided. Was any consideration given to also quantifying eye size?

We have processed the images to enhance their contrast and make the aberrations clearer. The percentage of degenerated eye area (Figure 1M) was normalized with total eye area. The method for quantifying degenerated area has been explained in the materials and methods section.

(3) Figure 1: a) Only enhancers of rough eyes are shown but no controls are included to evaluate whether knockdown of these genes causes eye toxicity in the absence of Tau. These are important missing controls. All putative Tau enhancers, including Pdn5/6, need to be tested with GMR-GAL4 independently of Tau to determine whether they cause a rough eye. In a previous publication from some of the same investigators (Raut et al 2017), knockdown of Pfdn using eyGAL4 was shown to induce severe eye morphology defects - this raises questions about the results shown here.

We agree that assessing the effects of HSP knockdown independent of Tau is essential to confirm modifier specificity. We have now performed these knockdowns, and the data are reported in Supplemental Table 1. For RNAi lines represented in Figure 1, which enhanced Tau-induced degeneration/eye developmental defect, except for one of the RNAi lines against Pfdn6 (GD34204), no detectable eye defects were observed when knocked down with GMR-Gal4 at 25°C, suggesting that enhancement is specific to the Tau background.

Use of a more eye-specific GMR-Gal4 driver at 25°C versus broader expressing ey-Gal4 at 29°C in prior work (Raut et al. 2017) likely reflects the differences in the eye morphological defects.

(b) Besides RNAi, do the classical Pdn5 deletion alleles included in this work also enhance the tau rough eye when heterozygous? Please also consider moving the Pfdn5/6 overexpression studies to evaluate possible suppression of the Tau rough eye to Figure 1, as it would enhance the interpretation of these data (but see also below).

GMR-Gal4 driven expression of hTau^{V337M} or hTau^{WT} in Pfdn5 heterozygous background does not enhance rough eye phenotype.

(4) For genes of special interest, such as Pdn5, and other genes mentioned in the results, the main figure, or discussion, it is also important to perform quantitative PCR to confirm that the RNAi lines used actually knock down mRNA expression and by how much. These studies will establish specificity.

We agree that confirming RNAi efficiency via quantitative PCR (qPCR) is essential for validating the knockdown efficiency. We have now included qPCR data, especially for key modifiers, confirming effective knockdown (Figure S2).

(5) Lines 235-238: how do you conclude whether the tau phenotype is "enhanced" when Pfdn5 causes a similar phenotype on its own? Could the combination simply be additive? Did overexpression of Pdn5 suppress the UAS-hTau NMJ bouton phenotype (see below)?

Although Pfdn5 mutants and hTau expression individually increase satellite boutons, their combination leads to a significantly more severe and additional phenotype, such as significantly decreased bouton size and increased bouton number, indicating an enhancing rather than purely additive interaction (Figure 4 and Figure S6C). Moreover, we now show

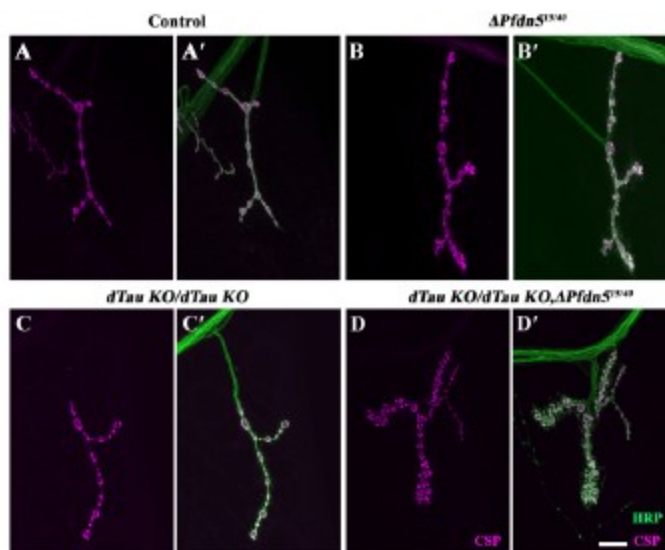
that overexpression of Pfdn5 significantly suppressed the hTau^{V337M}-induced NMJ phenotypes. This new data has been incorporated as Figure S11F-L in the revised manuscript.

Alternatively, did the authors consider reducing fly tau in the Pdn5 mutant background?

In new additional experiments, we observe that double mutants for *Drosophila* Tau (dTau) and Pfdn5 also exhibit severe NMJ defects, suggesting genetic interactions between dTau and Pfdn5. This data is shown below for the reviewer.

Author response image 1.

A double mutant combination of dTau and Pfdn5 aggravates the synaptic defects at the *Drosophila* NMJ. (A-D') Confocal images of NMJ synapses at muscle 4 of A2 hemisegment showing synaptic morphology in (A-A') control, (B-B') Δ Pfdn5^{15/40}, (C-C') dTauKO/dTauKO (*Drosophila* Tau mutant), (D-D') dTauKO/dTauKO; Δ Pfdn5^{15/40} double immunolabeled for HRP (green), and CSP (magenta). The scale bar in D for (A-D') represents 10 μ m.



(6) It may be important to further extend the investigation to the actin cytoskeleton. It is noted that Pfdn5 also stabilizes actin. Importantly, tau-mediated neurodegeneration in Drosophila also disrupts the actin cytoskeleton, and many other regulators of actin modify tau phenotypes.

We appreciate the suggestion to examine the actin cytoskeleton. While prior studies indicate that Pfdn5 might regulate the actin cytoskeleton and that Tau^{V377M} hyperstabilizes the actin cytoskeleton, we did not observe altered actin levels in Pfdn5 mutants (Figure 2G). However, actin dynamics may represent an additional mechanism through which Pfdn5 might temporally influence Tauopathy. Future work will address potential actin-related mechanisms in Tauopathy.

(7) Figure 2: in the provided images, it is difficult to appreciate the futsch loops. Please include an image with increased magnification. It appears that fly strains harboring a genomic rescue BAC construct are available for Pfdn-this would be a complementary reagent to test besides Pfdn overexpression.

We have updated Figure 2 to include high magnification NMJ images as insets, clearly showing the Futsch loops. While we have not yet tested a genomic rescue BAC construct for

Pfdn5, we plan to use the fly line harboring this construct in future work.

(8) Figure 3: Some of the data is not adequately explained. The use of Ran as a loading control seems rather unusual. What is the justification? Pfdn appears to only partially co-localize with α -tubulin in the axon; can the authors discuss or explain this? Further, in Pfdn5 mutants, there appears to be a loss of α -tubulin staining (3b); this should also be discussed.

We appreciate the reviewer's concern regarding the choice of loading control for our Western blot analysis. Importantly, since Tubulin levels and related pathways were the focus of our analysis, traditional loading controls such as α - or β -tubulin or actin were deemed unsuitable due to potential co-regulation. Ran, a nuclear GTPase involved in nucleocytoplasmic transport, is not known to be transcriptionally or post-translationally regulated by Tubulin-associated signaling pathways. To ensure its reliability as a loading control, we confirmed by densitometric analysis that Ran expression showed minimal variability across all samples. Hence, we used Ran for accurate normalization in the Western blot data represented in this manuscript. We have also used GAPDH as a loading control and found no difference with respect to Ran as a loading control across samples.

We appreciate the reviewer's comment regarding the interpretation of our Pearson's correlation coefficient (PCC) results. While the mean colocalization value of 0.6 represents a moderate positive correlation (MUKAKA 2012), which may not reach the conventional threshold for "high positive" colocalization (usually considered 0.7-0.9), it nonetheless indicates substantial spatial overlap between the proteins of interest. Importantly, colocalization analysis provides supportive but indirect evidence for molecular proximity. To further validate the interaction, we performed a microtubule binding assay, which directly demonstrates the binding of Pfdn5 to stabilized microtubules.

In accordance with the western blot analysis shown in Figure 2G-I, the levels of Tubulin are reduced in the Pfdn5 mutants (Figure 3B"). We have incorporated and discussed this in the revised manuscript.

(9) Figure 4: Overexpression of Pfdn appears to rescue the supernumerary satellite bouton numbers induced by human Tau; however, interpretation of this experiment is somewhat complicated as it is performed in Pfdn mutant genetic background. Can overexpression of Pfdn on its own rescue the Tau bouton defect in an otherwise wildtype background?

We have now coexpressed Pfdn5 and hTau^{V337M} in an otherwise wild-type background. As shown in Figure S11F-L, Pfdn5 overexpression suppresses Tau-induced bouton defects. We have incorporated the data in the Results section to support the role of Pfdn5 as a modifier of Tau toxicity.

(10) Lines 256-263 / Figure 5: (a) What exactly are these tau-positive structures (punctae) being stained in larval brains in Fig 5C-E? Most prior work on tau aggregation using Drosophila models has been done in the adult brain, and human wildtype or mutant Tau is not known to form significant numbers of aggregates in neurons (although aggregates have been described following glia tau expression).

Therefore, the results need to be further clarified. Besides the provided schematic, a zoomed-out image showing the whole larval brain is needed here for orientation. Have these aggregates been previously characterized in the literature?

We agree with the reviewer that the expression of the wildtype or mutant form of human Tau in Drosophila is not known to form aggregates in the larval brain, in contrast to the adult

brain (JACKSON et al. 2002; OKENVE-RAMOS et al. 2024). Consistent with previous reports, we also observed that Tau expression on its own does not form aggregates in the *Drosophila* larval brain.

However, in the absence of Pfdn5, microtubule disruption is severe, leading to reduced Tau-microtubule binding and formation of globular/round or flame-shaped tangles like aggregates in the larval brain. Previous studies have reported that 1,6-hexanediol can dissolve the Tau aggregate seeds formed by Tau droplets, but cannot dissolve the stable Tau aggregates (WEGMANN et al. 2018). We observed that 5% 1,6-Hexanediol failed to dissolve these Tau puncta, demonstrating the formation of stable aggregates in the absence of Pfdn5. Additionally, we now performed a Tau solubility assay and show that in the absence of Pfdn5, a significant amount of Tau goes in the pellet fraction, which could not be detected by phospho-specific AT8 Tau antibody (targeting pSer202/pThr205) but was detected by total hTau antibody (D5D8N) on the western blots (Figure S8). These data further reinforce our conclusion that Pfdn5 prevents the transition of hTau from soluble and/or microtubule-associated state to an aggregated, insoluble, and pathogenic state. These new data have been incorporated into the revised manuscript.

(b) Can additional markers (nuclei, cell membrane, etc.) be used to highlight whether the tau-positive structures are present in the cell body or at synapses?

We performed the co-staining of Tau and Elav to assess the aggregated Tau localization. We found that in the presence of Pfdn5, Tau is predominantly cytoplasmic and localised to the cell body and axons. In the absence of Pfdn5, Tau forms aggregates but is still localized to the cell body or axons. However, some of the aggregates are very large, and the subcellular localization could not be determined (Figure S8M-N'). These might represent brain regions of possible nuclear breakdown and cell death (JACKSON et al. 2002).

(c) It would also be helpful to perform western blots from larval (and adult) brains examining tau protein levels, phospho-tau species, possible higher-molecular weight oligomeric forms, and insoluble vs. soluble species. These studies would be especially important to help interpret the potential mechanisms of observed interactions.

Western blot analysis revealed that overexpression of Pfdn5 does not alter total Tau levels (Figure 6O). In Pfdn5 mutants, however, hTau^{V337M} levels were reduced in the supernatant fraction and increased in the pellet fraction, indicating a shift from soluble monomeric Tau to aggregated Tau.

(d) Does overexpression of Pdn5 (UAS-Pdn5) suppress the formation of tau aggregates? I would therefore recommend that additional experiments be performed looking at adult flies (perhaps in Pfdn5 heterozygotes or using RNAi due to the larval lethality of Pdn5 null animals).

Overexpression of Pfdn5 significantly reduced Tau-aggregates (Elav-Gal4/UAS-Tau^{V337M}; UAS-Pfdn5; DPfdn5^{15/40}) observed in Pfdn5 mutants (Figure 5E). Coexpression of Pfdn5 and hTau^{V337M} suppresses the Tau aggregates/puncta in 30-day adult brain. Since heterozygous DPfdn5^{15/+} did not show a reduction in Pfdn5 levels, we did not test the suppression of Tau aggregates in DPfdn5^{15/+}; Elav>UAS-Pfdn5, UAS-Tau^{V337M}.

(11) Figure 6, panels A-N: The GMR>Tau rough eye is not a "neurodegenerative" but rather a predominantly developmental phenotype. It results from aberrant retinal developmental patterning and the subsequent secretion/formation of the overlying eye cuticle (lenslets). I am confused by the data shown suggesting a "shrinking eye size" and increasing roughened surface over time (a GMR>tau eye similar to that shown in panel B

cannot change to appear like the one in panel H with aging). The rough eye can be quite variable among a population of animals, but it is usually fixed at the time the adult fly ecloses from the pupal case, and quite stable over time in an individual animal. Therefore, any suppression of the Tau rough eye seen at 30 days should be appreciable as soon as the animals eclose. These results need to be clarified. If indeed there is robust suppression of Tau rough eye, it may be more intuitive and clearer to include these data with Figure 1, when first showing the loss-of-function enhancement of the Tau rough eye. Also, why is Pfdn6 included in these experiments but not in the studies shown in Figures 2-5?

We thank the reviewer for their careful and knowledgeable assessment of the GMR>Tau rough eye model. We appreciate the clarification that the rough eye phenotype could be “developmental” rather than neurodegenerative.” Our initial observations regarding “shrinking eye size” and “increased surface roughness” clearly show age-related progression of structural change. Such progression has been observed and reported by others (IIJIMA-ANDO et al. 2012; PASSARELLA AND GOEDERT 2018). We observed an age-dependent increase in the number of fused ommatidia in GMR-Gal4 >Tau, which were rescued by Pfdn5 or Pfdn6 expression. We noted that adult-specific induction of hTau^{V337M} adult flies using the Gal80^{TS} and GMR-GeneSwitch (GMR-GS) systems was not sufficient to induce a significant eye phenotype; thus, early expression of Tau in the developing eye imaginal disc appears to be required for the adult progressive phenotype that we observe. We feel that it is inadequate to refer to this adult progressive phenotype as “developmental,” and while admittedly arguable whether this can be termed “degenerative.”

To address neurodegeneration more directly, we focused on 30-day-old adult fly brains and demonstrated that Pfdn5 overexpression suppresses age-dependent Tau-induced neurodegeneration in the central nervous system (Figure 6H-N and Figure S12). This supports our central conclusion regarding the neuroprotective role of Pfdn5 in age-associated Tau pathology. Since we found an enhancement in the Tau-induced synaptic and eye phenotypes by Pfdn6 knockdown, we also generated CRISPR/Cas9-mediated loss-of-function mutants for Pfdn6. However, loss of Pfdn6 resulted in embryonic/early first instar lethality, which precluded its detailed analysis at the larval stages.

(12) Figure 6, panels O-T: the elav>tau image appears to show a different frontal section plane compared to the other panels. It is advisable to show images at a similar level in all panels since vacuolar pathology can vary by region. It is also useful to be able to see the entire brain at a lower power, but the higher power inset view is obscuring these images. I would recommend creating separate panels rather than showing them as insets.

In the revised figure, we now display the low- and high-magnification images as separate, clearly labeled panels instead of using insets. This improves visibility of the brain morphology while providing detailed views of the vacuolar pathology (Figure 6H-L).

(13) Figure 6/7: For the experiments in which Pfdn5/6 is overexpressed and possibly suppresses tau phenotypes (brain vacuoles and memory), it is important to use controls that normalize the number of UAS binding sites, since increased UAS sites may dilute GAL4 and reduced Tau expression levels/toxicity. Therefore, it would be advisable to compare with Elav>Tau flies that also include a chromosome with an empty UAS site or other transgenes, such as UAS-GFP or UAS-lacZ.

We thank the reviewer for the suggestion. Now we have incorporated proper controls in the brain vacuolization, the mushroom body, and ommatidial fusion rescue experiments. Also,

we have independently verified whether Gal4 dilution has any effect on the Tau phenotypes (Figure 6H-L, Figure 7, and Figure S11A-B).

(14) Lines 311-312: the authors say vacuolization occurs in human neurodegenerative disease, which is not really true to my knowledge and definitely not stated in the citation they use. Please re-phrase.

Now we have made the appropriate changes in the revised manuscript.

(15) Figure 7: The authors claim that Pfdn5/6 expression does not impact memory behavior, but there in fact appears to be a decrease in preference index (panel D vs panel B). Does this result complicate the interpretation of the potential interaction with Tau (panel F). Are data from wildtype control flies available?

In our memory assay, a decrease in performance index (PI) of the trained flies compared to the naïve flies indicates memory formation (normal memory in control flies, Figure 7B). In contrast, a lack of significant difference in PI indicates a memory defect (Figure 7C: hTau^{V337M} overexpressed flies). "Decrease in preference index (panel D vs panel B)" is not a sign of memory defect; it may be interpreted as a better memory instead. Hence, neuronal overexpression of Pfdn5 (Figure 7D) or Pfdn6 (Figure 7E) in wildtype neurons does not cause memory deficits. In addition, coexpression of Pfdn5/6 and hTau^{V337M} successfully rescues the Tau-induced memory defect (significant drop in PI compared to the PI of naïve flies in Figure 7F-G). Moreover, almost complete rescue of the Tau-induced mushroom body defect on Pfdn5 or Pfdn6 expression further establishes potential interaction between Pfdn5/6 and Tau. This data has been incorporated into the revised manuscript.

The memory assay itself with extensive data on wildtype flies and various other genotype will shortly be submitted for publication in another manuscript (Majumder et al, manuscript under preparation); However, we can confirm for the reviewer that wildtype flies, trained and assayed by the protocol described, show a significant decrease in performance index compared to the naïve flies, indicative of strong learning and memory performance, very similar to the control genotype data shown in Figure 7B.

Additional minor considerations

(16) Lines 50-52: there are many therapeutic interventions for treating tauopathies, but not curative or particularly effective ones.

Now we have made the appropriate changes in the revised manuscript.

(17) Lines 87-106 seem like a duplication of the abstract. Consider deleting or condensing.

We have made the appropriate changes in the revised manuscript.

(18) Where is pfdn5 expressed? Development v. adult? Neuron v. glia? Conservation?

Prefoldin5 is expressed throughout development but strongly localized to the larval trachea and neuronal axons. Drosophila Pfdn5 shows 35% overall identity with human PFDN5.

(19) Liine 187: is pfdn5 truly "novel"?

The role of Pfdn5 as microtubule-binding and stabilizing is a new finding and has not been predicted or described before. Hence, it is a novel neuronal microtubule-associated protein.

(20) Figure 5, panel F, genotype labels on the x-axis are confusing; consider simplifying to Control, DPfdn, and Rescue.

We have made appropriate changes in the figure for better readability.

(21) Figures 5/8: it might be preferable to use consistent colors for Tau/HRP--Tau is labeled green in Figure 5 and then purple in Figure 8.

We have made these changes where possible.

(22) Lines 311-312: Vacuolar neuropathology is NOT typically observed in human Tauopathy.

We thank the reviewer for pointing this out. We have made the appropriate changes in the revised manuscript.

(23) Lines 328-349: The explanation could be made more clear. Naïve flies should not necessarily be called controls. Also, a more detailed explanation of how the preference index is computed would be helpful. Why are some datapoints negative values?

(a) We have rewritten this paragraph to make the description and explanation clearer. The detailed method and formula to calculate the Preference index have been incorporated in the Materials and Methods section.

(b) We have replaced the term Control with Naïve.

(c) Datapoints with negative values appeared in some of the 'Trained' group flies. It indicates that post-CuSO₄ training, some groups showed repulsion towards the otherwise attractive odor 2,3B. As 2,3B is an attractive odorant, naïve or control flies show attraction towards it compared to air, which is evident from a higher number of flies in the Odor arm (O) compared to that of the Air arm (A) of the Y-maze; thus, the PI $[(O-A)/(O+A)*100]$ is positive in case of naïve fly groups. Training of the flies led to an association of the attractive odorant with bitter food, leading to a decrease of attraction, and even repulsion towards the odorant in a few instances, resulting in less fly count in the odor arm compared to the air arm. Hence, the PI becomes negative as (O-A) is negative in such instances. Thus, it is not an anomaly but indicates strong learning.

(24) Line 403: misspelling "Pdfn"

We have corrected this.

(25) Lines 423-425: recommend re-phrasing, since tauopathies are human diseases. Mice and other animal models may be susceptible to tau-mediated neuronal dysfunction but not Tauopathy, per se.

We have made the appropriate changes in the revised manuscript.

(26) Lines 468-469: "tau neuropathology" rather than "tau associated neuropathies".

We have made the appropriate changes in the revised manuscript.

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